The orthoblique rotation for factor analysis

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For the last two decades, the gradient projection algorithm (GPA) has been the most used tool to rotate factor solutions. However, GPA is currently limited to performing either fully orthogonal or fully oblique rotations. Keeping structural zeros in selected positions of the factor correlation matrix after rotation is currently unavailable in Exploratory Factor Analysis. This is a shortcoming because in some cases it is known in advance that some factor correlations must be zero by theory or design. Thereby, we developed a new kind of rotation, the orthoblique rotation, in which both oblique and orthogonal relations between factors coexist. Building on the orthogonal and oblique algorithms of Jennrich, we provide a deeper account of how GPA works and use this account to generalize factor rotation to the orthoblique case. We then demonstrate that this rotation helps to estimate exploratory bifactor models with multiple general factors in which the general factors may be correlated but must remain orthogonal to the group factors, which can be considered a generalization of the biquartimin and bigeomin criteria of Jennrich and Bentler (2011; 2012). We also illustrate its application to exploratory multitrait-multimethod models, where trait and method factors may correlate within their respective sets but trait–method correlations are constrained to zero by design. Finally, we made available the new orthoblique rotation in the R package `latent`, whose `lrotate()` function serves the same purpose as the `GPArotation` package of Bernaards and Jennrich (2005). Taken together, this work contributes to filling the gap between the confirmatory–exploratory continuum and can be seen as a continuation of the efforts of Jennrich to broaden the flexibility and reach of Exploratory Factor Analysis.

Keywords: orthoblique rotation, exploratory factor analysis, gradient projection algorithm

Introduction

Exploratory factor analysis (EFA) is a popular framework for estimating large and complex factor structures. It surpasses the overly restrictive measurement models and the unacceptable model fit indices often encountered in applications of confirmatory factor analysis [CFA; Marsh et al. (2020)]. In CFA, incorrectly fixing cross-loadings to zero is an important caveat whereas EFA permits the estimation of all the loading parameters of the measurement model by means of a rotation step, reducing the risk of model misspecification.

However, the flexibility of EFA comes at the expense of specifying a suitable simplicity criterion to be optimized, so that a factor solution is rotated to a loading matrix that best approximates the desired simple structure. In this regard, the gradient projection algorithm [GPA; Jennrich (2001); Jennrich (2002)] has been the predominant computational tool for rotating loading patterns during the last twenty years. The success of GPA is instantiated by its availability in widely used and reputed statistical programs such as Mplus (Muthén & Muthén, 2017), `lavaan` (Rosseel, 2012), and `psych` (Revelle, 2020). In contrast with previously used rotation procedures like the criterion-specific pairwise algorithms discussed by Jennrich (2001), GPA is simple and elegant. In fact, a general-purpose optimization routine based on GPA can be written in about 50 lines of code in the R language [1]. Moreover, GPA is based on an optimization framework where traditional search directions based on first- and second-order derivatives and step-size selection techniques like backtracking line-search can be used (Liu, 2020).

The generality of GPA for factor rotation was the main contribution of Jennrich (2001) and Jennrich (2002). Rather than developing a different algorithm for every simplicity criterion, Jennrich showed that a common computational procedure could optimize many criteria under the same rotation constraints. Bernaards and Jennrich (2005) developed this idea into general-purpose software and separated the evaluation of the criterion and the gradient of the parameters from the operations that enforce the rotation constraints. They also provided derivations, proofs, and software implementation details. Thus, their work provided both the motivation and the computational starting point for the present generalization to a new rotation method. Our purpose is to give a deeper and more explicit geometric account of all the mathematical objects involving factor rotation and the connections between them. Explaining why they are needed, how they are derived from the constraints of the rotation

[1] See the `GPForth` and `GPFoblq` functions from the `GPArotation` R package (Bernaards & Jennrich, 2005) for orthogonal and oblique rotation, respectively

matrix, and how they work together provides a constructive route from the orthogonal and oblique cases to other rotation geometries.

This generalization is very important in EFA because the applications of GPA are currently limited to either orthogonal and oblique rotation. In other words, all the factors in an EFA model are currently constrained to be either orthogonal or allowed to correlate. This circumstance is problematic in cases where the researchers know in advance that two sets of factors may be correlated within each set but orthogonal between sets. For example, consider multitrait-multimethod models, in which trait and method factors may correlate within their respective sets but trait–method correlations must be constrained to zero by design. In this scenario, there should be structural zeros in the factor correlation matrix to effectively separate the variance explained by traits and methods. In other cases, it may be desirable to constrain one factor, or a group of factors, to be orthogonal to the remaining factors in order to obtain a more parsimonious pattern structure. For this scenario, consider applications of exploratory bifactor modeling with one or multiple general factors in which all the general factors must be orthogonal to the group factors, but may be correlated among themselves [Jiménez et al. (2023); jimenez_2023b]. If the general factors were allowed to correlate with the group factors in an oblique rotation analysis, then the variance explained by each kind of factor could not be correctly disentangled and would complicate the interpretation of the general-factor loadings.

Thereby, a generalization of the oblique and orthogonal rotations that allows for a mix of oblique and orthogonal factors could aid the correct estimation and interpretation of such factor structures. An illustrative example of this generalization is presented in Figure 1 along with the standard orthogonal and oblique cases.

Orthogonal

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Oblique

$$\begin{pmatrix} 1 & * & * & * & * & * \\ * & 1 & * & * & * & * \\ * & * & 1 & * & * & * \\ * & * & * & 1 & * & * \\ * & * & * & * & 1 & * \\ * & * & * & * & * & 1 \end{pmatrix}$$

Orthoblique

$$\left(\begin{array}{cc|cc|cc} 1 & * & 0 & 0 & 0 & 0 \\ * & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & * & 0 & 0 \\ 0 & 0 & * & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

Figure 1: Factor correlation matrices expected for six factors after orthogonal, oblique, and orthoblique rotations. Asterisks indicate freely estimated correlations.

In the orthoblique example presented in the figure, factors are organized into two oblique blocks of two factors each, with the remaining two factors being completely orthogonal. However, other correlation patterns are possible by modifying the size of the oblique and orthogonal blocks of factors. As such, the differential feature of the orthoblique rotation is that some blocks of factors may be allowed to correlate within but not between.

The objectives of this work are fourfold. First, we give a coherent and self-contained explanation of EFA and GPA for orthogonal and oblique rotation, connecting the matrix formulas of Jennrich (2001) and Jennrich (2002) with their underlying geometry. Second, using these explanations, we derive the orthoblique rotation, a generalization in which both oblique and orthogonal relations between factors coexist, and show how the conventional oblique and orthogonal rotations are obtained as special cases. We also discuss how to obtain standard errors for the rotated parameters. Third, we make this new rotation available in the `latent` R package and explain how the function `lrotate()` extends the rotation framework used in `GPArotation`. In addition, we provide the more general `lefa()` function to conduct EFA. Lastly, we illustrate the usefulness of this generalization in two empirical examples concerning exploratory multitrait-multimethod and bifactor models with multiple general factors. In the end, we discuss the benefits of the orthoblique rotation for preserving the flexibility of EFA to uncover cross-loadings and wrong theoretical expectations while retaining a desired factor-correlation pattern with structural zeros.

Factor rotation

In EFA, the covariance matrix $\boldsymbol{\Sigma}$ among the observed items is modeled as a function of the factor loading matrix $\boldsymbol{\Lambda}$, the factor correlation matrix $\boldsymbol{\Psi}$, and the error covariance matrix $\boldsymbol{\Theta}$. The loading matrix contains the relationships between the observed items and the common factors, the factor correlation matrix describes the associations among these factors, and the residual covariance matrix is assumed to be diagonal, containing the error variances or uniquenesses.

EFA is usually performed in two steps. First, an identified model $\boldsymbol{\Sigma} = \boldsymbol{\Lambda}_0\boldsymbol{\Lambda}_0^\top + \boldsymbol{\Theta}$ is estimated by maximum likelihood or least squares, where $\boldsymbol{\Lambda}_0$ is loading matrix constrained to have either orthogonal columns or a lower-triangular structure. Second, $\boldsymbol{\Lambda}_0$ is rotated to find an alternative but equally fitting model $\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Psi}\boldsymbol{\Lambda}^\top + \boldsymbol{\Theta}$, where $\boldsymbol{\Lambda} = \boldsymbol{\Lambda}_0\mathbf{X}^{-\top}$ and $\boldsymbol{\Psi} = \mathbf{X}^\top\mathbf{X}$.

This last step involves finding the optimal rotation matrix $\mathbf{X}$ that minimizes a complexity criterion seeking to obtain a sparse and interpretable $\mathbf{\Lambda}$. However, $\mathbf{X}$ must be subject to certain constraints that distinguish orthogonal from oblique rotations. In orthogonal rotation, the constraint is $\mathbf{X}^\top\mathbf{X} = \mathbf{I}$, whereas in oblique rotation it is $\text{diag}(\mathbf{X}^\top\mathbf{X}) = \mathbf{1}$. Therefore, both orthogonal and oblique rotation matrices belong to smooth nonlinear sets of $\mathbb{R}^{n\times n}$ matrices characterized by these specific constraints. Informally, these specific sets are termed manifolds and denoted by $\mathcal{M}$.

The problem of factor rotation is that the optimal orthogonal or oblique rotation matrix $\mathbf{X}$ cannot generally be found by using ordinary optimization algorithms that do not enforce these constraints. For this purpose, the Gradient Projection Algorithm (GPA) has proved to be an elegant and robust solution. GPA builds on differential geometry to perform optimization on nonlinear manifolds. Let $\tilde{f}$ be a complexity function with domain in $\mathbb{R}^{n\times n}$ that we aim to minimize with respect to a rotation matrix $\mathbf{X}$. This function can be any complexity criterion, such as varimax, oblimin, or geomin. Suppose that we want to constrain the solution to lie on a nonlinear manifold $\mathcal{M}$ characterized by a defining smooth function $h : \mathbb{R}^{n\times n} \to \mathbb{R}^d$ that satisfies

$$\mathcal{M} = h^{-1}(\mathbf{0}) = \left\{\mathbf{X} \in \mathbb{R}^{n\times n} : h(\mathbf{X}) = \mathbf{0}\right\},$$

so that any matrix $\mathbf{X} \in \mathcal{M}$ implies $h(\mathbf{X}) = \mathbf{0}$. Here, d is the number of independent constraints defining the manifold. This means that, in the orthogonal case, h collects the $n(n+1)/2$ distinct constraints of the symmetric matrix $\mathbf{X}^\top\mathbf{X}-\mathbf{I}$, whereas in the oblique case it collects the n diagonal constraints. In this manner, the orthogonal and oblique manifolds can be stated as the sets of full-rank $\mathbf{X}$ matrices satisfying the following conditions, respectively:

$$\begin{aligned}\mathcal{O}(n) &= \left\{\mathbf{X} \in \mathbb{R}^{n\times n} : h(\mathbf{X}) = \mathbf{X}^\top\mathbf{X} - \mathbf{I} = \mathbf{0}\right\},\\ \mathcal{OB}(n) &= \left\{\mathbf{X} \in \mathbb{R}^{n\times n} : h(\mathbf{X}) = \text{diag}(\mathbf{X}^\top\mathbf{X}) - \mathbf{1} = \mathbf{0}\right\}.\end{aligned}$$

To better understand these manifold definitions, an illustration of the orthogonal and oblique manifolds for $n = 3$ is presented in Figure 2. In both manifolds, each column $\mathbf{x}_j$ of the rotation matrix is a unit vector and can therefore be represented on the unit sphere. The oblique manifold is the product of n unit spheres, whereas the orthogonal manifold is the subset of this product in which the vectors are also constrained to point in mutually perpendicular directions.

Orthogonal rotation $\mathcal{O}(3)$ **Oblique rotation** $\mathcal{OB}(3)$

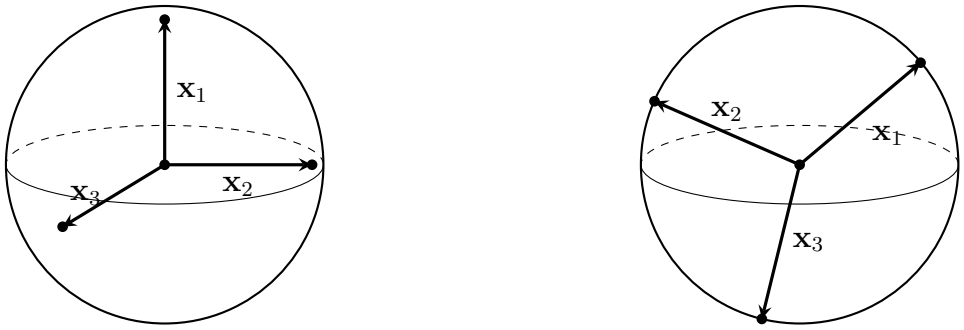


Figure 2: Schematic representation of the orthogonal and oblique manifolds in $n = 3$. In both cases, the columns of $\mathbf{X}$ are unit vectors on a sphere. In $\mathcal{O}(3)$ they are constrained to be orthogonal, whereas in $\mathcal{OB}(3)$ they may point in nonperpendicular directions.

Visually, the GPA consists of moving these vectors across the surface of the unit sphere until the complexity criterion $\tilde{f}$ is minimized, subject to the additional perpendicularity constraints in orthogonal rotation. To move the vectors in this way while minimizing $\tilde{f}$, small steps must be taken through an iterative algorithm. However, this is not trivial. At iteration k, traditional unconstrained optimization algorithms usually compute a new candidate as

$$\mathbf{X}_{k+1} = \mathbf{X}_k + t_k \mathbf{V}_k, \tag{1}$$

where $\mathbf{V}_k$ is a direction along which the complexity function decreases and t_k is a step size. In gradient descent, for example, $\mathbf{V}_k = -\text{grad}\tilde{f}(\mathbf{X}_k)$, where $\text{grad}\tilde{f}(\mathbf{X}_k)$ is the Euclidean gradient in the unrestricted space $\mathbb{R}^{n\times n}$.[2] The problem is that this updating scheme does not guarantee that the new candidate $\mathbf{X}_{k+1}$ remains on the manifold, not even when $\mathbf{X}_k$ already belongs to it. Furthermore, the descent direction $\mathbf{V}_k$ is also unrestricted in $\mathbb{R}^{n\times n}$ and is not necessarily a valid descent direction on a nonlinear manifold. In other words, a straight-line displacement in the Euclidean space, as shown in the updating Equation 1, is not generally a feasible displacement on a nonlinear manifold.

Hence, the GPA solves two problems: (1) proposing an appropriate descent direction $\mathbf{V}_k$ on $\mathcal{M}$ and (2) securing that, after moving along this direction, the new candidate $\mathbf{X}_{k+1}$ remains on $\mathcal{M}$.

[2]The Euclidean gradient is characterized by $\text{D}\tilde{f}(\mathbf{X}_k)[\mathbf{V}] = \langle \mathbf{V}, \text{grad}\tilde{f}(\mathbf{X}_k)\rangle$ for any $\mathbf{V} \in \mathbb{R}^{n\times n}$, with $\langle ., . \rangle$ denoting the Frobenius inner product.

Solution to problem 1: the Riemannian gradient

The Riemannian gradient is a generalization of the Euclidean gradient that represents first-order directions on a manifold. As such, it provides the direction towards the complexity criterion decreases when moving over nonlinear manifolds like the spheres depicted in Figure 2. More precisely, the Riemannian gradient is characterized as follows (Boumal, 2023, Definition 3.58):

Definition 1 *The Riemannian gradient of f at $\mathbf{X} \in \mathcal{M}$, denoted by $gradf(\mathbf{X})$, is the unique element of $\mathcal{T}_{\mathbf{X}}\mathcal{M}$ that satisfies $\mathrm{D}f(\mathbf{X})[\mathbf{V}] = \langle \mathbf{V}, gradf(\mathbf{X}) \rangle_{\mathbf{X}}$ for every $\mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M}$.*

Informally, if we zoom in enough around a point $\mathbf{X} \in \mathcal{M}$, eventually everything looks linear.[3] Thus, an idea to move over $\mathcal{M}$ consists of finding tangent directions to $\mathbf{X} \in \mathcal{M}$ that look linear from an infinitesimal point of view.

Consider doing a first-order Taylor expansion of h around the current candidate $\mathbf{X}_k$,

$$h(\mathbf{X}_k + t\mathbf{V}_k) \approx h(\mathbf{X}_k) + t\,\mathrm{D}h(\mathbf{X}_k)[\mathbf{V}_k]. \tag{2}$$

Because we can always propose a candidate $\mathbf{X}_k$ that already belongs to $\mathcal{M}$ [4], it follows that $h(\mathbf{X}_k) = \mathbf{0}$. Then, we may choose a $\mathbf{V}_k$ such that $\mathrm{D}h(\mathbf{X}_k)[\mathbf{V}_k] = \mathbf{0}$, so that Equation 2 becomes $\mathbf{0}$ and we preserve the manifold constraints to first order. Therefore, appropriate directions to move along the manifolds constitute the kernel of $\mathrm{D}h(\mathbf{X}_k)$:

$$\mathcal{T}_{\mathbf{X}}\mathcal{M} \equiv \ker \mathrm{D}h(\mathbf{X}) = \left\{ \mathbf{V} \in \mathbb{R}^{n \times n} : \mathrm{D}h(\mathbf{X})[\mathbf{V}] = \mathbf{0} \right\},$$

where the subscript k is dropped hereafter for clarity and generality. The set $\mathcal{T}_{\mathbf{X}}\mathcal{M}$ is known as the tangent space of $\mathbf{X}$ on $\mathcal{M}$. In other words, $\mathcal{T}_{\mathbf{X}}\mathcal{M}$ is the set of all infinitesimally feasible directions at $\mathbf{X}$ that are linear. This linearity property is important because it allows traditional optimization algorithms that work in $\mathbb{R}^{n \times n}$ to be used in the GPA framework.

Connecting this notion of linear space to our previous examples, the tangent spaces of the orthogonal and oblique manifolds at any $\mathbf{X} \in \mathcal{M}$ can be stated as the collection of directions

[3] As an analogy, the Earth is round but locally everything looks flat.
[4] We can always propose an initial starting point $\mathbf{X}_1 \in \mathcal{M}$.

for which the differentials of their respective defining functions are $\mathbf{0}$:

$$\begin{aligned}\mathcal{T}_{\mathbf{X}}\mathcal{O}(n) &= \left\{\mathbf{V} \in \mathbb{R}^{n\times n} : \mathbf{X}^\top\mathbf{V} + \mathbf{V}^\top\mathbf{X} = \mathbf{0}\right\}, \\ \mathcal{T}_{\mathbf{X}}\mathcal{OB}(n) &= \left\{\mathbf{V} \in \mathbb{R}^{n\times n} : \operatorname{ddiag}(\mathbf{X}^\top\mathbf{V}) = \mathbf{0}\right\},\end{aligned} \tag{3}$$

where ddiag is an operator that sets the off-diagonal entries of a matrix to zero.

Notice that tangent spaces can be viewed as a collection of the $n^2 - d$ free components in the manifolds, so that $\dim(\mathcal{T}_{\mathbf{X}}\mathcal{O}(n)) = n(n-1)/2$ is the number of free elements in an orthogonal matrix and $\dim(\mathcal{T}_{\mathbf{X}}\mathcal{OB}(n)) = n(n-1)$ is the number of free elements in an oblique matrix. Thereby, we can exploit the above differentials to parameterize the tangent spaces in such a way that the parameterizations preserve the tangent space dimension.

In the orthogonal case, for instance, if we set $\mathbf{V} = \mathbf{X}\boldsymbol{\Omega}$, then by substitution in Equation 3, we have

$$\mathcal{T}_{\mathbf{X}}\mathcal{O}(n) = \left\{\mathbf{X}\boldsymbol{\Omega} : \boldsymbol{\Omega} + \boldsymbol{\Omega}^\top = \mathbf{0}\right\}.$$

This is a valid full-rank parameterization of the tangent space of the orthogonal manifold because $\boldsymbol{\Omega}$ is a skew-symmetric matrix and hence it has $n(n-1)/2$ free elements.

In the oblique case, for a nonsingular rotation matrix $\mathbf{X}$, a useful full-rank parameterization of the tangent space is $\mathbf{V} = \mathbf{X}^{-\top}\boldsymbol{\Omega}$, which by substitution in Equation 3 yields

$$\mathcal{T}_{\mathbf{X}}\mathcal{OB}(n) = \left\{\mathbf{X}^{-\top}\boldsymbol{\Omega} : \operatorname{diag}(\boldsymbol{\Omega}) = \mathbf{0}\right\}.$$

Indeed, $\boldsymbol{\Omega}$ is an arbitrary off-diagonal matrix with $n(n-1)$ free elements and thus it is a valid full-rank parameterization of the tangent space of the oblique manifold.

Now, notice that any member of $\mathbb{R}^{n\times n}$ can be decomposed in orthogonal complements. At any point $\mathbf{X} \in \mathcal{M}$, the Euclidean space can be decomposed into the orthogonal direct sum of the tangent and normal spaces,

$$\mathbb{R}^{n\times n} = \mathcal{T}_{\mathbf{X}}\mathcal{M} \oplus \mathcal{N}_{\mathbf{X}}\mathcal{M}.$$

If this $\mathbb{R}^{n\times n}$ element is the Euclidean gradient of $\tilde{f}$ at $\mathbf{X}$, this means that it can be decomposed into an element belonging to the tangent space, $\mathcal{T}_{\mathbf{X}}\mathcal{M}$, and another belonging to the normal space, $\mathcal{N}_{\mathbf{X}}\mathcal{M}$. Therefore, there must exist orthogonal complements satisfying

$$\text{grad}\tilde{f}(\mathbf{X}) = \text{grad}\tilde{f}(\mathbf{X})_{\parallel} + \text{grad}\tilde{f}(\mathbf{X})_{\perp}, \tag{4}$$

where $\text{grad}\tilde{f}(\mathbf{X})_{\parallel}$ and $\text{grad}\tilde{f}(\mathbf{X})_{\perp}$ are the projections of $\text{grad}\tilde{f}(\mathbf{X})$ onto the tangent and normal spaces of the manifold $\mathcal{M}$, respectively (Absil et al., 2009, p. 47). Figure 3 depicts the relationship between these quantities.

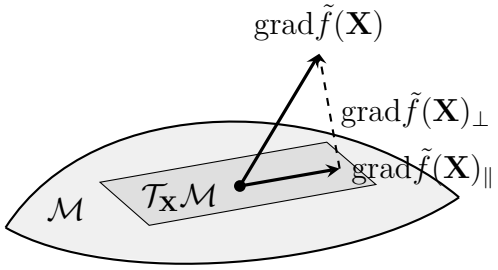


Figure 3: Manifold with the tangent space at $\mathbf{X}$, represented by the black dot. The tangent space is the flat rectangle at $\mathbf{X}$. It can be seen that the Euclidean gradient $\text{grad}\tilde{f}(\mathbf{X})$ can be decomposed in orthogonal complements $\text{grad}\tilde{f}(\mathbf{X})_{\perp}$ and $\text{grad}\tilde{f}(\mathbf{X})_{\parallel}$.

As it can be seen in Figure 3, a tangent space is indeed a linearization of the manifold at a given $\mathbf{X}$ and for any vector in the tangent space there would be another vector from the normal space that is orthogonal to it and satisfies Equation 4. However, to make sense of this notion of orthogonality at any $\mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M}$, we must endow the tangent space with a metric like the Frobenius inner product (i.e., $\langle \mathbf{U}, \mathbf{V} \rangle_{\mathbf{X}} = \text{trace}(\mathbf{U}^{\top}\mathbf{V})$), so that $\mathcal{M}$ acquires a Riemannian structure, turning it into a Riemannian manifold. In other words, the collection of inner products on the tangent spaces defines a Riemannian metric on $\mathcal{M}$, turning it into a Riemannian manifold, which allows us to develop a notion of orthogonal complements at each tangent space.

Let $f = \tilde{f}|_{\mathcal{M}}$ be the restriction of the complexity criterion $\tilde{f}$ to a manifold. Using Equation 4 and the Frobenius inner product as Riemannian metric, we arrive at the following identities

for any $\mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M}$:

$$\begin{aligned}
\left\langle \mathbf{V}, \mathrm{grad}\tilde{f}(\mathbf{X})_{\|} \right\rangle_{\mathbf{X}} &= \mathrm{trace}\left[\mathbf{V}^{\top}\left(\mathrm{grad}\tilde{f}(\mathbf{X}) - \mathrm{grad}\tilde{f}(\mathbf{X})_{\perp}\right)\right] \\
&= \mathrm{trace}\left(\mathbf{V}^{\top}\mathrm{grad}\tilde{f}(\mathbf{X})\right) - \mathrm{trace}\left(\mathbf{V}^{\top}\mathrm{grad}\tilde{f}(\mathbf{X})_{\perp}\right) \\
&= \mathrm{trace}\left(\mathbf{V}^{\top}\mathrm{grad}\tilde{f}(\mathbf{X})\right), \quad \text{because } \mathbf{V} \text{ is orthogonal to } \mathrm{grad}\tilde{f}(\mathbf{X})_{\perp}, \\
&= \mathrm{D}\tilde{f}(\mathbf{X})[\mathbf{V}], \quad \text{by the definition of the Euclidean gradient,} \\
&= \mathrm{D}f(\mathbf{X})[\mathbf{V}], \quad \text{because } f = \tilde{f}|_{\mathcal{M}} \text{ and } \mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M} \\
&= \langle \mathbf{V}, \mathrm{grad}f(\mathbf{X}) \rangle_{\mathbf{X}}.
\end{aligned}$$

The last equality follows directly from the definition of the Riemannian gradient (1). And from the preceding identities, this element turns out to be $\mathrm{grad}\tilde{f}(\mathbf{X})_{\|}$, the projection of the Euclidean gradient onto the tangent space of $\mathcal{M}$.

There can be many ways to find a formula for the Riemannian gradient, but a good strategy consists of obtaining an expression for $\mathrm{grad}\tilde{f}(\mathbf{X})_{\perp} \in \mathcal{N}_{\mathbf{X}}\mathcal{M}$ and substituting this into Equation 4.

In the cases of the orthogonal and oblique manifolds, recall that the parameterizations of the tangent spaces were $\mathbf{X}\boldsymbol{\Omega}$ and $\mathbf{X}^{-\top}\boldsymbol{\Omega}$, respectively, so their corresponding normal spaces must be

$$\begin{aligned}
\mathcal{N}_{\mathbf{X}}\mathcal{O}(n) &= \left\{\mathbf{N} : \mathbf{N} = \mathbf{X}\mathbf{A}, \quad \mathbf{A} = \mathbf{A}^{\top}\right\}, \\
\mathcal{N}_{\mathbf{X}}\mathcal{OB}(n) &= \left\{\mathbf{N} : \mathbf{N} = \mathbf{X}\mathbf{A}, \quad \mathbf{A} \text{ is diagonal}\right\}.
\end{aligned}$$

Clearly, the tangent and normal parameterizations are orthogonal in each scenario. In the orthogonal manifold, for any skew-symmetric $\boldsymbol{\Omega}$ and symmetric $\mathbf{A}$ we have $\langle \mathbf{X}\boldsymbol{\Omega}, \mathbf{X}\mathbf{A} \rangle = 0$. In the oblique manifold, for an off-diagonal $\boldsymbol{\Omega}$ and diagonal $\mathbf{A}$ we have $\langle \mathbf{X}^{-\top}\boldsymbol{\Omega}, \mathbf{X}\mathbf{A} \rangle = 0$. Therefore, in both cases the tangent and normal parameterizations are orthogonal complements and, using Equation 4, the Riemannian gradient can be written as

$$\mathrm{grad}f(\mathbf{X}) = \mathrm{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\mathbf{A}, \tag{5}$$

where $\mathbf{A}$ must be chosen so that $\mathrm{grad}f(\mathbf{X})$ belongs to the corresponding tangent space.

For orthogonal rotation, the Riemannian gradient must satisfy the tangent space equation

$$\mathbf{X}^{\top}\mathrm{grad}f(\mathbf{X}) + \mathrm{grad}f(\mathbf{X})^{\top}\mathbf{X} = \mathbf{0}.$$

Substituting $\text{grad}f(\mathbf{X})$ by the orthogonal complements parameterization in Equation 5 gives

$$\begin{aligned}
\mathbf{X}^\top\left(\text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\mathbf{A}\right) + \left(\text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\mathbf{A}\right)^\top\mathbf{X} &= \mathbf{0} \\
\mathbf{X}^\top\text{grad}\tilde{f}(\mathbf{X}) + \text{grad}\tilde{f}(\mathbf{X})^\top\mathbf{X} - \mathbf{X}^\top\mathbf{X}\mathbf{A} - \mathbf{A}^\top\mathbf{X}^\top\mathbf{X} &= \mathbf{0}, \quad \text{and} \\
\mathbf{X}^\top\text{grad}\tilde{f}(\mathbf{X}) + \text{grad}\tilde{f}(\mathbf{X})^\top\mathbf{X} - 2\mathbf{A} &= \mathbf{0}, \quad \text{because } \mathbf{X}^\top\mathbf{X} = \mathbf{I} \text{ and } \mathbf{A} = \mathbf{A}^\top.
\end{aligned}$$

It then follows that

$$\mathbf{A} = \frac{1}{2}\left[\mathbf{X}^\top\text{grad}\tilde{f}(\mathbf{X}) + \text{grad}\tilde{f}(\mathbf{X})^\top\mathbf{X}\right].$$

Finally, defining $\text{sym}(\mathbf{B}) = \frac{1}{2}\left(\mathbf{B} + \mathbf{B}^\top\right)$, we have $\mathbf{A} = \text{sym}\left(\mathbf{X}^\top\text{grad}\tilde{f}(\mathbf{X})\right)$ and the Riemannian gradient for orthogonal rotation becomes

$$\text{grad}f(\mathbf{X}) = \text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\,\text{sym}\left(\mathbf{X}^\top\text{grad}\tilde{f}(\mathbf{X})\right).$$

On the other hand, for oblique rotation, the Riemannian gradient must satisfy the tangent space equation

$$\text{ddiag}\left[\mathbf{X}^\top\text{grad}f(\mathbf{X})\right] = \mathbf{0}.$$

Substituting again $\text{grad}f(\mathbf{X})$ for the orthogonal complements parameterization in Equation 5 gives

$$\begin{aligned}
\text{ddiag}\left[\mathbf{X}^\top\left(\text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\mathbf{A}\right)\right] &= \mathbf{0} \\
\text{ddiag}\left[\mathbf{X}^\top\text{grad}\tilde{f}(\mathbf{X})\right] - \text{ddiag}\left(\mathbf{X}^\top\mathbf{X}\mathbf{A}\right) &= \mathbf{0}.
\end{aligned}$$

Because $\mathbf{A}$ is diagonal and $\text{ddiag}\left(\mathbf{X}^\top\mathbf{X}\right) = \mathbf{I}$, we get $\mathbf{A} = \text{ddiag}\left(\mathbf{X}^\top\mathbf{X}\mathbf{A}\right)$.

Therefore,

$$\mathbf{A} = \text{ddiag}\left[\mathbf{X}^\top\text{grad}\tilde{f}(\mathbf{X})\right],$$

and the Riemannian gradient for oblique rotation becomes

$$\text{grad}f(\mathbf{X}) = \text{grad}\tilde{f}(\mathbf{X}) - \mathbf{X}\text{ddiag}\left[\mathbf{X}^\top\text{grad}\tilde{f}(\mathbf{X})\right].$$

These are indeed the same projected-gradient formulas denoted by `Gp` in the orthogonal and oblique implementations of Bernaards and Jennrich (pp. 692–695; 2005) and in the

`GPArotation` package.

Solution to problem 2: the Retraction

With this generalization of gradients to nonlinear manifolds, we are prepared to compute valid descent directions along the tangent space $\mathcal{T}_{\mathbf{X}}\mathcal{M}$. However, we are not ready to compute valid candidates because the update

$$\mathbf{Z}_{k+1} = \mathbf{X}_k - t_k \text{grad} f(\mathbf{X}_k) \tag{6}$$

does not generally belong to $\mathcal{M}$, however small t_k is. The tangent direction only provides a first-order linear approximation to the manifold at $\mathbf{X}_k$, as we showed before in Equation 2. This means that the condition $\text{D}h(\mathbf{X})[\mathbf{V}] = \mathbf{0}$ does not generally imply that the tangent step $\mathbf{Z} = \mathbf{X} + t\mathbf{V}$ lies exactly on $\mathcal{M}$ for a nonzero t. It only ensures that the constraints are preserved to first order. To see this, let's take the crossproduct of the tangent step $\mathbf{Z}$ to check how it departs from the manifold constraints:

$$\begin{aligned}\mathbf{Z}^\top\mathbf{Z} &= (\mathbf{X} + t\mathbf{V})^\top(\mathbf{X} + t\mathbf{V}) \\ &= \mathbf{X}^\top\mathbf{X} + t(\mathbf{X}^\top\mathbf{V} + \mathbf{V}^\top\mathbf{X}) + t^2\mathbf{V}^\top\mathbf{V}.\end{aligned}$$

In orthogonal rotation, the term $\mathbf{X}^\top\mathbf{V} + \mathbf{V}^\top\mathbf{X}$ vanishes because $\mathbf{V}$ was chosen precisely to satisfy the tangent equation for the orthogonal manifold (Equation 3) and we are left with

$$\mathbf{Z}^\top\mathbf{Z} = \mathbf{I} + t^2\mathbf{V}^\top\mathbf{V}. \tag{7}$$

This means that with the updating scheme in Equation 6, the orthogonality constraint is violated by the residual $t^2\mathbf{V}^\top\mathbf{V}$. Following the same argument in the case of the oblique manifold, we have

$$\text{ddiag}(\mathbf{Z}^\top\mathbf{Z}) = \mathbf{I} + t^2\text{ddiag}(\mathbf{V}^\top\mathbf{V}), \tag{8}$$

and the unit norm constraint would be off by a factor of $t^2\text{ddiag}(\mathbf{V}^\top\mathbf{V})$.

Mapping the tangent step back to the manifold is the second problem addressed by the GPA. After taking a step t along the negative Riemannian gradient, the resulting point must

be projected back onto the manifold while preserving the direction of the tangent step. For this purpose, suppose that we apply some projection $R_{\mathbf{X}} : \mathcal{T}_{\mathbf{X}}\mathcal{M} \to \mathcal{M}$ that restores the constraints exactly while preserving this direction. Because the tangent step is already correct to first order, $R_{\mathbf{X}}$ only needs to act on the higher-order residuals of eq-cross_step_orth and Equation 8. This is exactly what a retraction does: it fixes the constraints of the tangent step $\mathbf{Z}$ without modifying the descent direction (i.e., the negative of the Riemannian gradient).

A retraction only needs to satisfy two conditions. The first condition ensures that taking a zero step leaves the current point unchanged. In other words, if we do not move $\mathbf{X}$ along any direction, then the projection must return the same $\mathbf{X}$. The second condition recognizes that the tangent step $\mathbf{X} + t\mathbf{V}$ already satisfies the manifold constraints to first order because $\mathbf{V}$ was picked to satisfy the tangent space equations. Therefore it requires that $R_{\mathbf{X}}(t\mathbf{V})$ has the same initial direction $\mathbf{V}$ as the tangent step. In other words, a retraction restores the manifold constraints exactly without changing the first-order direction on the manifold so that the Riemannian gradient is still the descent direction.

Using these conditions, we can define a retraction in the following manner (Boumal, 2023, Definition 3.47):

Definition 2 *A retraction is a smooth map $R_{\mathbf{X}} : \mathcal{T}_{\mathbf{X}}\mathcal{M} \to \mathcal{M}$ that satisfies the following conditions:*
(1) $R_{\mathbf{X}}(\mathbf{0}) = \mathbf{X}$, and
(2) $DR_{\mathbf{X}}(\mathbf{0})[\mathbf{V}] = \mathbf{V}$ for every $\mathbf{V} \in \mathcal{T}_{\mathbf{X}}\mathcal{M}$.

Combining the Riemannian gradient with a retraction, the valid updating equation that GPA uses becomes

$$\mathbf{X}_{k+1} = R_{\mathbf{X}_k}\left(-t_k \text{grad} f(\mathbf{X}_k)\right). \tag{9}$$

Hence, the Riemannian gradient determines a valid descent direction in the tangent space, whereas the retraction guarantees that the new candidate $\mathbf{X}_{k+1}$ belongs to $\mathcal{M}$ after correcting the higher-order imprecisions of the tangent step. Figure 4 gives a notion of how a tangent step is retracted onto the manifold.

For orthogonal rotation, a suitable retraction is obtained by applying the Gram-Schmidt algorithm to the columns of $\mathbf{Z}$ such that $R^{\text{orth}}_{\mathbf{X}}(t\mathbf{V}) = \mathbf{Q}$, where $\mathbf{Q}$ is the orthonormal matrix of the QR decomposition of the tangent step $\mathbf{Z}$, with the triangular factor chosen to have positive diagonal entries. Alternatively, if $\mathbf{Z} = \mathbf{U}\mathbf{D}\mathbf{W}^\top$ is a singular value decomposition,

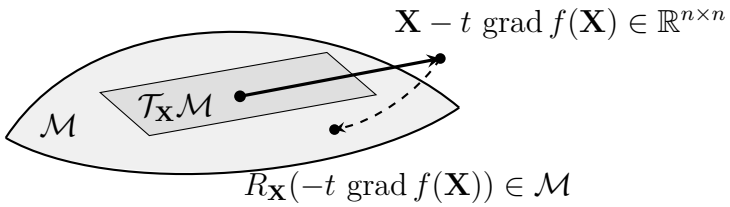


Figure 4: Illustration of a retraction. Starting from $\mathbf{X} \in \mathcal{M}$, a tangent step is taken along the negative Riemannian gradient in the tangent space $\mathcal{T}_{\mathbf{X}}\mathcal{M}$. Since the point $\mathbf{X} - t \text{ grad } f(\mathbf{X})$ does not generally belong to the manifold, the retraction maps it back onto $\mathcal{M}$.

another valid retraction is $R_{\mathbf{X}}^{\text{orth}}(t\mathbf{V}) = \mathbf{U}\mathbf{W}^\top$. This is actually the orthogonal projection used in the algorithms of Jennrich (2001) and Bernaards and Jennrich (2005). The QR and singular value decomposition retractions need not produce the same finite update, but both are smooth operations that satisfy the conditions in Definition 2, leaving the point $\mathbf{X}$ unchanged when no direction is applied and preserving the tangent direction to first order.

Let's define the retraction for orthogonal rotation as

$$R_{\mathbf{X}}^{\text{orth}}(t\mathbf{V}) = \mathbf{Z}\left(\mathbf{Z}^\top\mathbf{Z}\right)^{-1/2}, \tag{10}$$

which is equivalent to the singular value decomposition method.

The first condition of Definition 2 is met because at $t = 0$, we have $R_{\mathbf{X}}^{\text{orth}}(\mathbf{0}) = \mathbf{X}$ as $\mathbf{X}$ is already orthonormal. To check the second condition, we can rewrite the retraction in terms of Equation 7 as follows:

$$R_{\mathbf{X}}^{\text{orth}}(t\mathbf{V}) = (\mathbf{X} + t\mathbf{V})\left(\mathbf{I} + t^2\mathbf{V}^\top\mathbf{V}\right)^{-1/2}.$$

Then the derivative of this expression with respect to t becomes

$$\frac{d}{dt}R_{\mathbf{X}}^{\text{orth}}(t\mathbf{V}) = \mathbf{V}\left(\mathbf{I} + t^2\mathbf{V}^\top\mathbf{V}\right)^{-1/2} - t(\mathbf{X} + t\mathbf{V})\left(\mathbf{I} + t^2\mathbf{V}^\top\mathbf{V}\right)^{-3/2}\mathbf{V}^\top\mathbf{V},$$

and evaluated at $t = 0$, we obtain $\frac{d}{dt}R_{\mathbf{X}}^{\text{orth}}(t\mathbf{V})\Big|_{t=0} = \mathbf{V}$ so that the descent direction along the tangent space is preserved.

In case of oblique rotation, a suitable retraction involves normalizing the columns of $\mathbf{Z}$:

$$R_{\mathbf{X}}^{\text{oblq}}(t\mathbf{V}) = \mathbf{Z}\left[\text{ddiag}\Big(\mathbf{Z}^\top\mathbf{Z}\Big)\right]^{-1/2}, \tag{11}$$

which is the projection used by Jennrich (2002, Appendix A.1).

At $t = 0$ we have $R_{\mathbf{X}}^{\text{oblq}}(\mathbf{0}) = \mathbf{X}$ because the columns of $\mathbf{X}$ are already unit norm. To prove the second condition, we rewrite the normalization using Equation 8 such that

$$R_{\mathbf{X}}^{\text{oblq}}(t\mathbf{V}) = (\mathbf{X} + t\mathbf{V})\left(\mathbf{I} + t^2\,\text{ddiag}(\mathbf{V}^\top\mathbf{V})\right)^{-1/2}.$$

Then the derivative with respect to t is

$$\frac{d}{dt}R_{\mathbf{X}}^{\text{oblq}}(t\mathbf{V}) = \mathbf{V}\left(\mathbf{I} + t^2\,\text{ddiag}(\mathbf{V}^\top\mathbf{V})\right)^{-1/2} - t(\mathbf{X}+t\mathbf{V})\left(\mathbf{I} + t^2\,\text{ddiag}(\mathbf{V}^\top\mathbf{V})\right)^{-3/2}\text{ddiag}(\mathbf{V}^\top\mathbf{V}),$$

which evaluated at $t = 0$ becomes $\frac{d}{dt}R_{\mathbf{X}}^{\text{oblq}}(t\mathbf{V})\Big|_{t=0} = \mathbf{V}$.

Using the definitions of Riemannian gradient and retraction along with their proofs for the orthogonal and oblique cases, we are ready to develop a new rotation: the orthoblique rotation.

The orthoblique manifold

The orthoblique manifold encompasses a mixture of oblique and orthogonal relations between factors. Geometrically, in $n = 3$ it consists of solving the same problem of orthogonal and oblique rotations by minimizing a complexity function while moving vectors on a sphere, with the difference that only one vector is required to be orthogonal to the others (Figure 5).

For clarity, we first consider the case of two blocks of oblique factors and another block of completely orthogonal factors. However, the following results can be generalized to any number of oblique blocks of factors.

Let $\mathcal{POB}\Big(\{p, q\}, r\Big)$ denote the set of orthoblique matrices in $\mathbb{R}^{n\times n}$ defined by

Orthoblique rotation

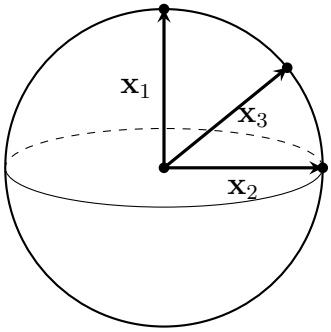


Figure 5: Schematic representation of the orthoblique manifold in $n = 3$. The columns of $\mathbf{X}$ are again unit vectors on a sphere. Here, $\mathbf{x}_1$ is orthogonal to $\mathbf{x}_2$ and $\mathbf{x}_3$, whereas $\mathbf{x}_2$ and $\mathbf{x}_3$ are oblique.

$$\mathcal{POB}\Big(\{p,q\},r\Big) = \left\{ \mathbf{X} \in \mathbb{R}^{n\times n} : \mathbf{X}^\top\mathbf{X} = \begin{bmatrix} \mathbf{Y}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{Y}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix}, \mathrm{diag}(\mathbf{Y}_1) = \mathbf{1}, \mathrm{diag}(\mathbf{Y}_2) = \mathbf{1} \right\}, \quad (12)$$

where p is the number of oblique factors in the first block, q is the number of oblique factors in the second block, r is the number of completely orthogonal factors in the third block, and thus $n = p + q + r$. Let's partition the rotation matrix accordingly as $\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 & \mathbf{X}_2 & \mathbf{X}_3 \end{bmatrix}$, where $\mathbf{X}_1 \in \mathbb{R}^{n\times p}$, $\mathbf{X}_2 \in \mathbb{R}^{n\times q}$, and $\mathbf{X}_3 \in \mathbb{R}^{n\times r}$. Then,

$$\mathbf{Y}_1 = \mathbf{X}_1^\top\mathbf{X}_1, \qquad \mathbf{Y}_2 = \mathbf{X}_2^\top\mathbf{X}_2, \qquad \mathbf{I} = \mathbf{X}_3^\top\mathbf{X}_3.$$

Thus, the columns in the oblique blocks $\mathbf{X}_1$ and $\mathbf{X}_2$ are oblique within but orthogonal in between whereas the columns in $\mathbf{X}_3$ are orthogonal both to each other and to the columns in the oblique blocks.

The workflow to derive the Riemannian gradient of the orthoblique manifold is as follows: First, we find a smooth defining function h to describe the manifold. Second, we develop a full-rank representation of the tangent and normal spaces in terms of $\mathbf{X}$. Third, using these parameterizations of the tangent and normal spaces, we find an expression for the projection of the Euclidean gradient of $\tilde{f}$ onto the tangent space. Fourth, we propose a retraction satisfying Definition 2.

Defining function

We can write the orthoblique manifold in terms of a defining function h that collects the constraints in block-matrix form such that $\mathcal{POB}\Big(\{p,q\},r\Big) = \{\mathbf{X} \in \mathbb{R}^{n\times n} : h(\mathbf{X}) = \mathbf{0}\}$, with

$$h(\mathbf{X}) = \begin{bmatrix} \operatorname{ddiag}(\mathbf{X}_1^\top \mathbf{X}_1) - \mathbf{I} & \mathbf{X}_1^\top \mathbf{X}_2 & \mathbf{X}_1^\top \mathbf{X}_3 \\ \mathbf{X}_2^\top \mathbf{X}_1 & \operatorname{ddiag}(\mathbf{X}_2^\top \mathbf{X}_2) - \mathbf{I} & \mathbf{X}_2^\top \mathbf{X}_3 \\ \mathbf{X}_3^\top \mathbf{X}_1 & \mathbf{X}_3^\top \mathbf{X}_2 & \mathbf{X}_3^\top \mathbf{X}_3 - \mathbf{I} \end{bmatrix}.$$

As already explained, h is understood to collect only the independent entries of this symmetric block matrix: the p and q diagonal constraints in the first two blocks, the $pq + pr + qr$ entries in the three upper off-diagonal blocks, and the $r(r+1)/2$ distinct entries of the third diagonal block. Hence, the number of independent constraints is $d = p + q + pq + pr + qr + r(r+1)/2$.

Tangent space

The tangent space is obtained by differentiating the previous block-matrix representation of $\mathcal{POB}\Big(\{p,q\},r\Big)$. First, let $\mathbf{V}$ be a tangent matrix with partitions $\mathbf{V} = \begin{bmatrix} \mathbf{V}_1 & \mathbf{V}_2 & \mathbf{V}_3 \end{bmatrix}$. Then, the tangent space can be represented as

$$\mathcal{T}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big) = \left\{ \mathbf{V} \in \mathbb{R}^{n\times n} : \begin{bmatrix} \operatorname{ddiag}(\mathbf{X}_1^\top \mathbf{V}_1) & \mathbf{X}_1^\top \mathbf{V}_2 + \mathbf{V}_1^\top \mathbf{X}_2 & \mathbf{X}_1^\top \mathbf{V}_3 + \mathbf{V}_1^\top \mathbf{X}_3 \\ \mathbf{X}_2^\top \mathbf{V}_1 + \mathbf{V}_2^\top \mathbf{X}_1 & \operatorname{ddiag}(\mathbf{X}_2^\top \mathbf{V}_2) & \mathbf{X}_2^\top \mathbf{V}_3 + \mathbf{V}_2^\top \mathbf{X}_3 \\ \mathbf{X}_3^\top \mathbf{V}_1 + \mathbf{V}_3^\top \mathbf{X}_1 & \mathbf{X}_3^\top \mathbf{V}_2 + \mathbf{V}_3^\top \mathbf{X}_2 & \mathbf{X}_3^\top \mathbf{V}_3 + \mathbf{V}_3^\top \mathbf{X}_3 \end{bmatrix} = \mathbf{0} \right\}. \tag{13}$$

At a nonsingular rotation matrix $\mathbf{X}$, we can parameterize this tangent space as $\mathbf{V} = \mathbf{X}^{-\top}\mathbf{\Omega}$, where $\mathbf{\Omega} = \mathbf{X}^\top \mathbf{V}$. Then, by substitution we have

$$\mathcal{T}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big) = \left\{ \mathbf{X}^{-\top}\mathbf{\Omega} : \begin{bmatrix} \operatorname{ddiag}(\mathbf{\Omega}_{11}) & \mathbf{\Omega}_{12} + \mathbf{\Omega}_{21}^\top & \mathbf{\Omega}_{13} + \mathbf{\Omega}_{31}^\top \\ \mathbf{\Omega}_{21} + \mathbf{\Omega}_{12}^\top & \operatorname{ddiag}(\mathbf{\Omega}_{22}) & \mathbf{\Omega}_{23} + \mathbf{\Omega}_{32}^\top \\ \mathbf{\Omega}_{31} + \mathbf{\Omega}_{13}^\top & \mathbf{\Omega}_{32} + \mathbf{\Omega}_{23}^\top & \mathbf{\Omega}_{33} + \mathbf{\Omega}_{33}^\top \end{bmatrix} = \mathbf{0} \right\},$$

where $\mathbf{\Omega}$ has the following block structure:

$$\mathbf{\Omega} = \begin{bmatrix} \mathbf{\Omega}_{11} & \mathbf{\Omega}_{12} & \mathbf{\Omega}_{13} \\ -\mathbf{\Omega}_{12}^\top & \mathbf{\Omega}_{22} & \mathbf{\Omega}_{23} \\ -\mathbf{\Omega}_{13}^\top & -\mathbf{\Omega}_{23}^\top & \mathbf{\Omega}_{33} \end{bmatrix}.$$

$\mathbf{\Omega}_{11}$ and $\mathbf{\Omega}_{22}$ can be arbitrary zero-diagonal matrices and $\mathbf{\Omega}_{33}$ must be skew-symmetric. The block matrices in the upper diagonal can be totally arbitrary, but they must be skew-symmetric with respect to the lower diagonal blocks. We can see that $\dim(\mathbf{\Omega}) = p(p-1) + q(q-1) + pq + pr + qr + r(r-1)/2$. Indeed, this is the same as $n^2 - d$ and hence $\mathbf{X}^{-\top}\mathbf{\Omega}$ is a valid full-rank parameterization of the tangent space.

Normal space

Following the same principle as for the orthogonal and oblique manifolds, the normal space is the orthogonal complement of the tangent space. A valid parameterization can be constructed by stacking the orthogonal complements of the individual blocks of $\mathbf{\Omega}$. We already showed that any diagonal matrix is an orthogonal complement of any off-diagonal matrix and that any symmetric matrix is an orthogonal complement of any skew-symmetric matrix. Therefore,

$$\mathcal{N}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big) = \left\{ \mathbf{N} = \mathbf{XA} : \mathbf{A} = \begin{bmatrix} \mathbf{D}_1 & \mathbf{B}_{12} & \mathbf{B}_{13} \\ \mathbf{B}_{12}^\top & \mathbf{D}_2 & \mathbf{B}_{23} \\ \mathbf{B}_{13}^\top & \mathbf{B}_{23}^\top & \mathbf{S} \end{bmatrix} \right\},$$

where $\mathbf{D}_1$ and $\mathbf{D}_2$ are diagonal matrices, $\mathbf{S}$ is symmetric, and $\mathbf{B}_{12}$, $\mathbf{B}_{13}$, and $\mathbf{B}_{23}$ are unrestricted matrices with transposed counterparts below the diagonal.

Indeed, if $\mathbf{V} = \mathbf{X}^{-\top}\mathbf{\Omega} \in \mathcal{T}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big)$ and $\mathbf{N} = \mathbf{XA} \in \mathcal{N}_{\mathbf{X}}\mathcal{POB}\Big(\{p,q\},r\Big)$, then

$$\begin{aligned} \langle \mathbf{V}, \mathbf{N} \rangle &= \operatorname{trace}\Big(\mathbf{V}^\top \mathbf{N}\Big) \\ &= \operatorname{trace}\Big(\mathbf{\Omega}^\top \mathbf{X}^{-1}\mathbf{XA}\Big) \\ &= \operatorname{trace}\Big(\mathbf{\Omega}^\top \mathbf{A}\Big) \\ &= 0, \end{aligned}$$

and the next step is to find the right $\mathbf{A}$ to construct an expression of the Riemannian gradient.

Riemannian gradient

For clarity, let $\mathbf{G} = \text{grad}\tilde{f}(\mathbf{X})$ denote the Euclidean gradient. As in the orthogonal and oblique cases, the Riemannian gradient here is obtained by subtracting its normal component (Equation 5), leaving $\text{grad}f(\mathbf{X}) = \mathbf{G} - \mathbf{X}\mathbf{A}$. The matrix $\mathbf{A}$ must be chosen so that $\text{grad}f(\mathbf{X}) \in \mathcal{T}_{\mathbf{X}}\mathcal{POB}$. Hence, substituting $\mathbf{V}$ for this parameterization of the Riemannian gradient in the equation for the tangent space, all the components in the following symmetric matrix become $\mathbf{0}$:

$$\begin{bmatrix} \text{ddiag}\big(\mathbf{X}_1^\top\mathbf{G}_1 - \mathbf{Y}_1\mathbf{D}_1\big) & \mathbf{X}_1^\top\mathbf{G}_2 + \mathbf{G}_1^\top\mathbf{X}_2 - \mathbf{Y}_1\mathbf{B}_{12} - \mathbf{B}_{12}\mathbf{Y}_2 & \mathbf{X}_1^\top\mathbf{G}_3 + \mathbf{G}_1^\top\mathbf{X}_3 - \mathbf{Y}_1\mathbf{B}_{13} - \mathbf{B}_{13} \\ & \text{ddiag}\big(\mathbf{X}_2^\top\mathbf{G}_2 - \mathbf{Y}_2\mathbf{D}_2\big) & \mathbf{X}_2^\top\mathbf{G}_3 + \mathbf{G}_2^\top\mathbf{X}_3 - \mathbf{Y}_2\mathbf{B}_{23} - \mathbf{B}_{23} \\ & & \mathbf{X}_3^\top\mathbf{G}_3 + \mathbf{G}_3^\top\mathbf{X}_3 - 2\mathbf{S} \end{bmatrix}.$$

Each block can now be solved separately.[5] Furthermore, the diagonal blocks of the above expression are the same equations as in the oblique and orthogonal demonstrations but represented in block format. Thereby, the unique $\mathbf{A}$ that satisfies Equation 5 becomes

$$\mathbf{A} = \begin{bmatrix} \mathbf{D}_1 = \text{ddiag}\big(\mathbf{X}_1^\top\mathbf{G}_1\big) & \mathbf{B}_{12} & \mathbf{B}_{13} \\ \mathbf{B}_{12}^\top & \mathbf{D}_2 = \text{ddiag}\big(\mathbf{X}_2^\top\mathbf{G}_2\big) & \mathbf{B}_{23} \\ \mathbf{B}_{13}^\top & \mathbf{B}_{23}^\top & \mathbf{S} = \text{sym}(\mathbf{X}_3^\top\mathbf{G}_3) \end{bmatrix}.$$

For the submatrices at the intersections of the oblique and orthogonal blocks, the solutions are

$$\mathbf{B}_{13} = (\mathbf{Y}_1 + \mathbf{I})^{-1}(\mathbf{X}_1^\top\mathbf{G}_3 + \mathbf{G}_1^\top\mathbf{X}_3),$$
$$\mathbf{B}_{23} = (\mathbf{Y}_2 + \mathbf{I})^{-1}(\mathbf{X}_2^\top\mathbf{G}_3 + \mathbf{G}_2^\top\mathbf{X}_3).$$

Finally, the only remaining block is the intersection between the two oblique blocks, where

$$\mathbf{Y}_1\mathbf{B}_{12} + \mathbf{B}_{12}\mathbf{Y}_2 = \mathbf{X}_1^\top\mathbf{G}_2 + \mathbf{G}_1^\top\mathbf{X}_2.$$

This is a Sylvester equation. Let $\mathbf{Q} = \mathbf{X}_1^\top\mathbf{G}_2 + \mathbf{G}_1^\top\mathbf{X}_2$, $\mathbf{Y}_1 = \mathbf{U}_1\mathbf{V}_1\mathbf{U}_1^\top$, and $\mathbf{Y}_2 = \mathbf{U}_2\mathbf{V}_2\mathbf{U}_2^\top$, where the last two expressions are the eigendecompositions of the symmetric correlation matrices $\mathbf{Y}_1$ and $\mathbf{Y}_2$, respectively. Then, by substitution we have

[5] Notice that we omitted the lower triangle of the matrix because it is redundant with the transpose of the upper triangle.

$$(\mathbf{U}_1\mathbf{V}_1\mathbf{U}_1^\top)\mathbf{B}_{12} + \mathbf{B}_{12}(\mathbf{U}_2\mathbf{V}_2\mathbf{U}_2^\top) = \mathbf{Q}.$$

Using the orthonormality property of the eigenvectors, multiplying the equation on the left by $\mathbf{U}_1^\top$ and on the right by $\mathbf{U}_2$, we get

$$\mathbf{V}_1(\mathbf{U}_1^\top\mathbf{B}_{12}\mathbf{U}_2) + (\mathbf{U}_1^\top\mathbf{B}_{12}\mathbf{U}_2)\mathbf{V}_2 = \mathbf{U}_1^\top\mathbf{Q}\mathbf{U}_2.$$

Let $\mathbf{W} = \mathbf{U}_1^\top\mathbf{B}_{12}\mathbf{U}_2$ so that the expression above can be written as $\mathbf{V}_1\mathbf{W} + \mathbf{W}\mathbf{V}_2 = \mathbf{U}_1^\top\mathbf{Q}\mathbf{U}_2$. Because $\mathbf{V}_1$ and $\mathbf{V}_2$ are diagonal, this equation can be solved elementwise as

$$w_{ij} = \frac{(\mathbf{U}_1^\top\mathbf{Q}\mathbf{U}_2)_{ij}}{(\mathbf{V}_1)_{ii} + (\mathbf{V}_2)_{jj}}.$$

Finally, multiplying again the left and right sides by $\mathbf{U}_1$ and $\mathbf{U}_2^\top$, respectively, the solution becomes

$$\mathbf{B}_{12} = \mathbf{U}_1\mathbf{W}\mathbf{U}_2^\top.$$

Retraction

After computing the Riemannian gradient, the remaining problem is to map $\mathbf{Z} = \mathbf{X} - t\mathrm{grad}f(\mathbf{X})$ back onto the orthoblique manifold. For this, we need to find a function $R_{\mathbf{X}}^{\mathcal{POB}} : \mathcal{T}_{\mathbf{X}}\mathcal{POB} \to \mathcal{POB}$ that satisfies the conditions of Definition 2.

Let $\mathbf{R}$ be the retracted tangent step $\mathbf{R} = R_{\mathbf{X}}^{\mathcal{POB}}(t\mathbf{V})$. Let us partition the matrices $\mathbf{Z}$ and $\mathbf{R}$ into blocks $\mathbf{Z} = \begin{bmatrix}\mathbf{Z}_1 & \mathbf{Z}_2 & \mathbf{Z}_3\end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2 & \mathbf{R}_3\end{bmatrix}$ such that the submatrices indexed by 1 and 2 correspond to the first two oblique blocks, respectively, and those indexed by 3 correspond to the block of orthogonal factors.

The retraction of the first oblique block can be simply the retraction of $\mathbf{Z}_1$ onto the space set of normalized columns (Equation 11),

$$\mathbf{R}_1 = \mathbf{Z}_1\left[\mathrm{ddiag}\left(\mathbf{Z}_1^\top\mathbf{Z}_1\right)\right]^{-1/2},$$

so that $\mathrm{diag}(\mathbf{R}_1^\top\mathbf{R}_1) = \mathbf{1}$.

The second oblique block also has to be made orthogonal to the first block. To do this, we first project $\mathbf{Z}_2$ onto the orthogonal space of $\mathbf{Z}_1$ before normalizing the result. First, define

the orthogonal projector onto the column space of $\mathbf{R}_1$ as $\mathbf{P}_1 = \mathbf{R}_1 \left(\mathbf{R}_1^\top \mathbf{R}_1\right)^{-1} \mathbf{R}_1^\top$. Then, $\widetilde{\mathbf{Z}}_2 = (\mathbf{I} - \mathbf{P}_1)\, \mathbf{Z}_2$ is orthogonal to $\mathbf{R}_1$. Second, the columns are subsequently normalized to obtain

$$\mathbf{R}_2 = \widetilde{\mathbf{Z}}_2 \left[\operatorname{ddiag}\left(\widetilde{\mathbf{Z}}_2^\top \widetilde{\mathbf{Z}}_2\right)\right]^{-1/2}.$$

Consequently, $\mathbf{R}_1^\top \mathbf{R}_2 = \mathbf{0}$ and $\operatorname{diag}\left(\mathbf{R}_2^\top \mathbf{R}_2\right) = \mathbf{1}$.

Finally, to get $\mathbf{R}_3$, combine the two oblique blocks columnwise in $\mathbf{R}_{12} = \begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2\end{bmatrix}$ and define the projector onto their joint column space, $\mathbf{P}_{12} = \mathbf{R}_{12} \left(\mathbf{R}_{12}^\top \mathbf{R}_{12}\right)^{-1} \mathbf{R}_{12}^\top$. As before, the orthogonal block is first projected onto the orthogonal complement of both oblique blocks to yield $\widetilde{\mathbf{Z}}_3 = (\mathbf{I} - \mathbf{P}_{12})\, \mathbf{Z}_3$ and its columns are then orthonormalized as in Equation 10 using

$$\mathbf{R}_3 = \widetilde{\mathbf{Z}}_3 \left(\widetilde{\mathbf{Z}}_3^\top \widetilde{\mathbf{Z}}_3\right)^{-1/2}$$

As a result, $\mathbf{R}_1^\top \mathbf{R}_3 = \mathbf{0}$, $\mathbf{R}_2^\top \mathbf{R}_3 = \mathbf{0}$, $\mathbf{R}_3^\top \mathbf{R}_3 = \mathbf{I}$ and $\mathbf{R}_3$ becomes completely orthogonal.

Taken together, the following blockwise construction defines the retracted orthoblique update

$$R_{\mathbf{X}}^{\mathcal{POB}}(t\mathbf{V}) = \begin{bmatrix}\mathbf{R}_1 & \mathbf{R}_2 & \mathbf{R}_3\end{bmatrix}. \tag{14}$$

Notice that we already proved that the orthonormalization (10) and normalization (11) operations are valid retractions for the orthogonal and oblique manifolds, respectively. To show that $R_{\mathbf{X}}^{\mathcal{POB}}$ is another valid retraction for the orthoblique manifold, we just need to show that the projection of rotation blocks onto the orthogonal space of previous rotation blocks also satisfies the retraction conditions.

First, define $P_{\mathbf{X}_j}$ to be a function that takes a block j and projects it onto the orthogonal space of a previously retracted block i:

$$\begin{aligned} P_{\mathbf{X}_j}(t\mathbf{V}_j) &= (\mathbf{I} - \mathbf{R}_i(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\mathbf{R}_i^\top)\mathbf{Z}_j \\ &= \mathbf{Z}_j - \mathbf{R}_i(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\mathbf{R}_i^\top \mathbf{Z}_j. \end{aligned} \tag{15}$$

At $t = 0$, $\mathbf{R}_i$ reduces to $\mathbf{X}_i$ and $\mathbf{Z}_j$ reduces to $\mathbf{X}_j$. Since $\mathbf{X}_i^\top \mathbf{X}_j = \mathbf{0}$, the second term of this expression vanishes and we are left with $P_{\mathbf{X}_j}(t\mathbf{V}_j)\Big|_{t=0} = \mathbf{X}_j$. Therefore, $R_{\mathbf{X}}^{\mathcal{POB}}(\mathbf{0}) = \mathbf{X}$, satisfying the first retraction condition.

To prove the second retraction condition we first differentiate Equation 15 using the product

rule,

$$\begin{aligned} DP_{\mathbf{X}_j}(t\mathbf{V}_j) = \mathbf{V}_j &- (D\mathbf{R}_i)(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\mathbf{R}_i^\top \mathbf{Z}_j \\ &- \mathbf{R}_i D\left[(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\right]\mathbf{R}_i^\top \mathbf{Z}_j \\ &- \mathbf{R}_i(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}(D\mathbf{R}_i)^\top \mathbf{Z}_j \\ &- \mathbf{R}_i(\mathbf{R}_i^\top \mathbf{R}_i)^{-1}\mathbf{R}_i^\top \mathbf{V}_j. \end{aligned}$$

Next, we evaluate this expression at $t = 0$. Again, since $\mathbf{R}_i$ reduces to $\mathbf{X}_i$ and $\mathbf{Z}_j$ reduces to $\mathbf{X}_j$, $\mathbf{X}_i^\top \mathbf{X}_j = \mathbf{0}$, the two terms containing $\mathbf{R}_i^\top \mathbf{Z}_j$ vanish. Therefore,

$$DP_{\mathbf{X}_j}(t\mathbf{V}_j)\Big|_{t=0} = \mathbf{V}_j - \mathbf{X}_i(\mathbf{X}_i^\top \mathbf{X}_i)^{-1}\left[D\mathbf{R}_i^\top\Big|_{t=0}\mathbf{X}_j + \mathbf{X}_i^\top \mathbf{V}_j\right].$$

If the previous projected block $\mathbf{R}_i$ already satisfies the second retraction condition, then $D\mathbf{R}_i|_{t=0} = \mathbf{V}_i$ and we obtain

$$DP_{\mathbf{X}_j}(t\mathbf{V}_j)\Big|_{t=0} = \mathbf{V}_j - \mathbf{X}_i(\mathbf{X}_i^\top \mathbf{X}_i)^{-1}\left(\mathbf{V}_i^\top \mathbf{X}_j + \mathbf{X}_i^\top \mathbf{V}_j\right).$$

But the tangent-space equations (13) state that $\mathbf{V}_i^\top \mathbf{X}_j + \mathbf{X}_i^\top \mathbf{V}_j = \mathbf{0}$. Hence, $DP_{\mathbf{X}_j}(t\mathbf{V}_j)\Big|_{t=0} = \mathbf{V}_j$.

Finally, because we already showed that doing normalization or orthogonalization preserve first-order directions, it follows that applying these operations afterwards still preserves this direction and hence $DR_{\mathbf{X}}^{\mathcal{POB}}(t\mathbf{V})\Big|_{t=0} = \mathbf{V}$, which proves the validity of our retraction (Equation 14).

Notice that any block order also satisfies these retraction conditions, provided each block is projected onto the orthogonal complement of all previously processed blocks and then normalized if they are oblique or orthonormalized if it is the orthogonal block. Different orders may produce different finite updates but they always preserve the same first-order direction.

Generalization to an arbitrary number of oblique blocks

The previous construction of the orthoblique manifold can be extended to any number m of oblique blocks. Let p_i be the number of factors in the ith oblique block and let r be the number of completely orthogonal factors, so that $n = \sum_{i=1}^{m} p_i + r$. Then,

$$\mathcal{POB}\Big(\{p_1,\ldots,p_m\},r\Big)=\left\{\mathbf{X}\in\mathbb{R}^{n\times n}:\mathbf{X}^\top\mathbf{X}=\begin{bmatrix}\mathbf{Y}_1 & \mathbf{0} & \ldots & \mathbf{0}\\ \vdots & \ddots & \mathbf{0} & \vdots\\ \vdots & \mathbf{0} & \mathbf{Y}_m & \mathbf{0}\\ \mathbf{0} & \ldots & \mathbf{0} & \mathbf{I}\end{bmatrix},\quad \text{diag}(\mathbf{Y}_i)=\mathbf{1}\ \forall i\right\}.$$

In this setup, the tangent space, normal space, and retraction would involve additional oblique blocks, pairs of oblique blocks, and pairs of oblique and orthogonal blocks. But all the formulas of these quantities follow by applying the same blockwise approach as before.

Furthermore, this generalization shows that orthogonal and oblique rotations are special cases of the orthoblique manifold. If there are no oblique blocks and $r = n$, then $\mathcal{POB}(\varnothing, n) = \mathcal{O}(n)$. And if there is one oblique block with $p_1 = n$ and $r = 0$, then $\mathcal{POB}(\{n\}, 0) = \mathcal{OB}(n)$.

Rotation constraints and standard errors

We already know how to impose constraints on $\mathbf{X}$ in orthogonal, oblique, and orthoblique rotation scenarios. A remaining matter is how these constraints are translated to the rotated loadings $\mathbf{\Lambda}$ and factor correlation matrix $\mathbf{\Psi}$ as this information is required for computing valid standard errors of rotated parameters.

Let θ be a vector collecting the k factor model parameters after factor rotation and $\mathbf{I}_\theta$ be the corresponding observed information matrix (i.e., the negative Hessian of the log-likelihood). Define $\mathbf{C}_\theta$ as the $k \times c$ matrix whose columns contain the derivatives of the c constraints with respect to these parameters that must be satisfied at the stationary point. Then,

$$\begin{pmatrix}\mathbf{I}_\theta & \mathbf{C}_\theta\\ \mathbf{C}_\theta^\top & \mathbf{0}\end{pmatrix} \tag{16}$$

is an augmented Information matrix whose inverse can be used to obtain a variance-covariance estimate in maximum likelihood estimation [Equation 17; Jennrich (1974)]. For non-maximum-likelihood estimators, an alternative augmented Hessian matrix can be incorporated into the sandwich covariance estimator described by Zhang (2014). Replacing $\mathbf{I}_\theta$ by the Hessian matrix of the loss function in Equation 16 would suffice.

For orthogonal and oblique rotations, the derivatives of the constraints can be found

elsewhere (Archer & Jennrich, 1973; Jennrich, 1973). For orthoblique rotation, we first derive the constraints on the rotated parameters and then compute their corresponding derivatives with respect to all the model parameters.

First, let ζ be the function that transforms the rotation matrix into the rotated parameters. In orthoblique rotation, $\zeta(\mathbf{X}) = \{\mathbf{\Lambda} = \mathbf{\Lambda}_0\mathbf{X}^{-\top}, \mathbf{\Psi} = \mathbf{X}^\top\mathbf{X}\}$. By the definition of gradient, at any tangent direction $\mathbf{V}$ we should have

$$\begin{aligned} \mathrm{D}f(\mathbf{X})[\mathbf{V}] &= \langle \Delta f_{\mathbf{\Lambda\Psi}}, \mathrm{D}\zeta(\mathbf{X})[\mathbf{V}] \rangle, \\ &= 0, \quad \text{at convergence}, \end{aligned} \tag{17}$$

where $\Delta f_{\mathbf{\Lambda\Psi}}$ is the gradient of the complexity function with respect to the rotated parameters. The second equality follows because at a given solution the complexity criterion must be stationary.

As shown before, the tangent space of the orthoblique manifold has the form $\mathbf{V} = \mathbf{X}^{-\top}\mathbf{\Omega}$, where the blocks of oblique factors in $\mathbf{\Omega}$ are arbitrary off-diagonal matrices, the block of orthogonal factors is skew-symmetric, and the intersections in $\mathbf{\Omega}$ between different blocks of factors i and j satisfy $\mathbf{\Omega}_{ij} = -\mathbf{\Omega}_{ji}^\top$.

Substituting this tangent parameterization into the differential of ζ gives

$$\begin{aligned} \mathrm{D}\zeta(\mathbf{X})[\mathbf{V}] &= \{-\mathbf{\Lambda}_0\mathbf{X}^{-\top}\mathbf{V}^\top\mathbf{X}^{-\top}, \mathbf{X}^\top\mathbf{V} + \mathbf{V}^\top\mathbf{X}\} \\ &= \{-\mathbf{\Lambda}(\mathbf{X}^{-\top}\mathbf{\Omega})^\top\mathbf{X}^{-\top}, \mathbf{X}^\top(\mathbf{X}^{-\top}\mathbf{\Omega}) + (\mathbf{X}^{-\top}\mathbf{\Omega})^\top\mathbf{X}\} \\ &= \{-\mathbf{\Lambda}\mathbf{\Omega}^\top\mathbf{\Psi}^{-1}, \mathbf{\Omega} + \mathbf{\Omega}^\top\}. \end{aligned}$$

At the stationary solution, substituting these derivatives into Equation 17 then gives

$$\begin{aligned} 0 &= \langle \Delta f_{\mathbf{\Lambda}}, -\mathbf{\Lambda}\mathbf{\Omega}^\top\mathbf{\Psi}^{-1} \rangle + \langle \Delta f_{\mathbf{\Psi}}, \mathbf{\Omega} + \mathbf{\Omega}^\top \rangle, \\ &= \mathrm{trace}(-\Delta f_{\mathbf{\Lambda}}^\top\mathbf{\Lambda}\mathbf{\Omega}^\top\mathbf{\Psi}^{-1}) + \mathrm{trace}(\Delta f_{\mathbf{\Psi}}^\top(\mathbf{\Omega} + \mathbf{\Omega}^\top)), \\ &= \mathrm{trace}(-\mathbf{\Psi}^{-1}\Delta f_{\mathbf{\Lambda}}^\top\mathbf{\Lambda}\mathbf{\Omega}^\top) + \mathrm{trace}(2\Delta f_{\mathbf{\Psi}}\mathbf{\Omega}^\top). \end{aligned}$$

Therefore, the orthoblique constraints imply that the matrix

$$\mathbf{K} = 2\Delta f_{\mathbf{\Psi}} - \mathbf{\Psi}^{-1}\Delta f_{\mathbf{\Lambda}}^\top\mathbf{\Lambda} \tag{18}$$

must be symmetric, with the submatrices corresponding to the blocks of oblique factors being

also required to be diagonal.

Following the orthoblique definition in (Equation 12), if p is the number of oblique factors in the first block, q is the number of oblique factors in the second block, and r is the number of completely orthogonal factors in the third block, then the number of stationarity equations is the dimension of the tangent space,

$$c = p(p-1) + q(q-1) + pq + pr + qr + r(r-1)/2.$$

The first two terms count the off-diagonal entries of each oblique block. The next three terms count the symmetry equations between different blocks, and the final term counts the symmetry equations within the orthogonal block. These are c stationarity equations that become zero at convergence and whose derivatives with respect to the rotated parameters are collected in the $\mathbf{C}_\theta$ term of the augmented Information representation (Equation 16).

Differentiating Equation 18 yields

$$\begin{aligned} \mathrm{d}\mathbf{K} = {} & 2\,\mathrm{d}(\Delta f_{\boldsymbol{\Psi}}) \\ & + \boldsymbol{\Psi}^{-1}(\mathrm{d}\boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}\Delta f_{\boldsymbol{\Lambda}}^{\top}\boldsymbol{\Lambda} \\ & - \boldsymbol{\Psi}^{-1}\left[\mathrm{d}(\Delta f_{\boldsymbol{\Lambda}})^{\top}\boldsymbol{\Lambda} + \Delta f_{\boldsymbol{\Lambda}}^{\top}\mathrm{d}\boldsymbol{\Lambda}\right]. \end{aligned} \tag{19}$$

To obtain the appropriate $k \times c$ derivatives in $\mathbf{C}_\theta$ from this differential, let $\mathbf{e}_i \in \mathbb{R}^k$ denote the ith standard basis vector, whose ith element is one and whose remaining elements are zero. For each model parameter $i = 1, \ldots, k$ in θ, we set the direction $\mathrm{d}\theta = \mathbf{e}_i$ in turn and evaluate Equation 19 along $\mathrm{d}\theta$ to find the first $p(p-1) + q(q-1)$ constraint derivatives corresponding to the off-diagonal entries of the oblique blocks. To find the remaining $pq + pr + qr + r(r-1)/2$ constraint derivatives of the symmetric entries corresponding to the intersections between blocks and the entries in the orthogonal block, simply evaluate $\mathrm{d}\mathbf{K} - \mathrm{d}\mathbf{K}^{\top}$ in the same iterative fashion.

Software: orthoblique rotation with the latent package

A general rotation framework should allow researchers to compare and use different factor structures without developing a new optimization routine for each application. The

`GPArotation` package already provides this framework for conventional orthogonal and oblique rotations, with a wide catalogue of rotation criteria. These are features retained in our `latent` package (Jiménez et al., 2026), with some additional capabilities. The relevant innovation here is the addition of the orthoblique manifold to restrict the factor-correlation structure as desired and the application of different rotation criteria to different blocks of factors.

Rotation in the `latent` package is performed through the `lrotate()` function, which can receive an initial loading matrix and factor correlation matrix to be rotated. An example is presented below:

```
library(latent)
# Orthogonal factor rotation with varimax:
fit_rotation <- lrotate(lambda = unrotated_lambda,
                        projection = "orth",
                        rotation = "varimax")
```

The choice of criterion is separate from the choice of geometry: `rotation` specifies the criterion, while `projection = "orth"` requests an orthogonal rotation. The default, `projection = "oblq"`, allows all factors to be oblique.

To choose orthoblique rotation we must use the `oblique` argument. For example, consider the following orthoblique target rotation:

```
library(latent)
# Orthoblique target rotation:
fit_rotation <- lrotate(lambda = unrotated_lambda,
                        oblique = c(3, 3),
                        rotation = "target", target = target)
```

Here, `oblique = c(3, 3)` automatically selects the orthoblique geometry and partitions the factors into two blocks of three factors each. As result, clumns 1–3 form the first oblique block, and columns 4–6 form the second. Correlations are allowed within each block, while correlations between blocks are constrained to zero. With six factors, the column ordering

and the correlation structure of the rotated solution become

$$\boldsymbol{\Lambda} = \left[\begin{array}{ccc|ccc} \underbrace{\boldsymbol{\lambda}_1 & \boldsymbol{\lambda}_2 & \boldsymbol{\lambda}_3}_{\text{oblique block 1}} & \multicolumn{3}{c}{\underbrace{\boldsymbol{\lambda}_4 \; \boldsymbol{\lambda}_5 \; \boldsymbol{\lambda}_6}_{\text{oblique block 2}}} \end{array}\right],$$

$$\boldsymbol{\Psi} = \left[\begin{array}{ccc|ccc} 1 & \psi_{12} & \psi_{13} & 0 & 0 & 0 \\ \psi_{12} & 1 & \psi_{23} & 0 & 0 & 0 \\ \psi_{13} & \psi_{23} & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 & \psi_{45} & \psi_{46} \\ 0 & 0 & 0 & \psi_{45} & 1 & \psi_{56} \\ 0 & 0 & 0 & \psi_{46} & \psi_{56} & 1 \end{array}\right].$$

The factor correlations are freely estimated within each oblique block, whereas correlations between blocks are constrained to zero. In this setup, there are many alternative choices for the `oblique` argument. The `oblique` argument accepts any vector of positive integers specifying the sizes of consecutive oblique blocks, which may vary in number and size as long as their total does not exceed the number of factors in the unrotated loading matrix. For example, with six factors, `oblique = c(2, 2, 2)` specifies three oblique blocks of two factors each, whereas `oblique = c(3, 2)` specifies two oblique blocks containing three and two factors, respectively. In the latter case, the remaining factor is assumed to be completely orthogonal. More generally, whenever the sum of the block sizes is smaller than the total number of factors, all remaining factors are constrained to be completely orthogonal.

As an alternative to `lrotate()`, the function `lefa()` for exploratory factor analysis takes the raw data or a covariance matrix as inputs and performs factor extraction first and rotation later. `lefa()` allows all the same arguments as `lrotate()` but contains additional choices for factor extraction with either maximum likelihood (`"ml"`), unweighted least-squares (`"uls"`), or diagonally-weighted least-squares (`"dwls"`).

```
# Oblique EFA with Maximum Likelihood and oblique oblimin rotation:
fit_efa <- lefa(data = data.matrix,
                sample.cov = NULL, # Optional covariance matrix
                nfactors = 6L,
                estimator = "ml",
                projection = "oblq",
```

```
                rotation = "oblimin")
```

Because different factor blocks may contain different levels of complexity, both `lrotate` and `lefa()` allow the application of different rotation criteria to arbitrary columns of $\mathbf{\Lambda}$. Suppose that we have a block of three oblique factors and another block of three orthogonal factors. Then, we may choose to apply oblimin to the oblique block to maximize item simplicity but varimax to the orthogonal block to maximize the variance of the squared loadings within factors. This rotation specification can be easily implemented by providing a list of rotation criteria to the `rotation` argument:

```
# Orthoblique EFA with combined oblimin + varimax rotations:
fit_efa <- lefa(data = data.matrix,
                nfactors = 6L,
                estimator = "ml",
                oblique = 3,
                rotation = list(
                  oblimin = list(factors = 1:3),
                  varimax = list(factors = 4:6)))
```

The value `oblique = 3` automatically selects the orthoblique manifold and allows correlations only among the first three factors. The remaining three factors are assumed to be completely orthogonal. Independently, the `rotation` argument applies oblimin to the first three loading columns and varimax to the last three columns. Thus, `oblique` determines the permitted factor-correlation structure, whereas `rotation` specifies the complexity criterion applied to each loading block. This particular combination of oblimin for oblique factors and varimax for orthogonal factors is intended solely to illustrate the flexibility of the rotation framework in the `lrotate` function and is not a recommendation for use in practice.

Another advantage of `lefa()` is that it computes the variance-covariance matrix and standard errors of the rotated parameters and uniquenesses using the corresponding rotation constraints derivatives.

Finally, `latent` can also estimate factor scores that preserve the orthoblique factor correlations. Currently, two existing methods implemented in the function `latscores` allow this: the constrained regression method of Green (1969) that finds the best correlation-preserving scores in the least-squares sense, and the method of ten Berge et al. (1999) that

minimizes the determinant of the residual covariance matrix:

```
# Compute the best factor scores in the least-squares sense:
Green_scores <- latscores(fit, method = "Green")
# Compute the factor scores that minimize the determinant of the residual
# covariance matrix:
tenBerge_scores <- latscores(fit, method = "tenBerge")
```

The development version of `latent` is available at https://github.com/Marcosjnez/latent, including the source and documentation of the functions `lrotate()`, `lefa()`, and `latscores()`. The R package can be installed with the command `pak("marcosjnez/latent")` that uses the `pak` library (Csárdi & Hester, 2026).

Alternative approaches towards an orthoblique structure

In this section we briefly discuss two alternative methods to orthoblique rotation and explain why orthoblique rotation is more general and has some advantages over them.

First, one may argue that achieving an orthoblique correlation pattern is already possible in Exploratory Structural Equation Modeling [ESEM; Asparouhov & Muthén (2009)]. In ESEM, rotation is usually presented as an optimization problem with independent exploratory blocks of factors that can be either oblique or orthogonal. Take as an example the orthoblique pattern presented in Figure 1. In the ESEM framework, we can reproduce the same correlation pattern by building a block-diagonal matrix with three square blocks of independent rotation matrices of size 2×2 each:

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{X}_3 \end{pmatrix}.$$

If $\mathbf{X}_1$ and $\mathbf{X}_2$ are subjected to oblique rotation and $\mathbf{X}_3$ to orthogonal rotation, then an orthoblique pattern is recovered for the factor correlation matrix:

$$\boldsymbol{\Psi} = \mathbf{X}^\top \mathbf{X} = \left(\begin{array}{cc|cc|cc} 1 & * & 0 & 0 & 0 & 0 \\ * & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & * & 0 & 0 \\ 0 & 0 & * & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right).$$

However, in this scenario the number of estimated parameters is smaller than in the more general orthoblique rotation framework presented here. Orthoblique rotation finds a dense rotation matrix $\mathbf{X}$ that yields the desired orthoblique pattern whereas ESEM solves a more restricted optimization problem with structural zeros in the intersections of different exploratory blocks of factors. Consequently, orthoblique rotation searches over a larger set of rotation matrices and can attain a lower value of the rotation criterion than the more restrictive block-diagonal approach took in ESEM. In these regards, orthoblique rotation is a more exploratory approach than ESEM.

Second, we also highlight that using orthoblique rotation in tandem with target rotation is closely related to the extended target rotation proposed by Zhang et al. (2019). Their method incorporates expectations about both the loading matrix and the factor correlation matrix into the rotation criterion, so that selected trait-method correlations are rotated toward zero. More precisely, the extended target rotation minimizes the following criterion using oblique rotation:

$$\frac{1}{2}\|\mathbf{W}_\Lambda \odot (\boldsymbol{\Lambda} - \mathbf{T}_\Lambda)\|^2 + \frac{w}{4}\|\mathbf{W}_\Psi \odot (\boldsymbol{\Psi} - \mathbf{T}_\Psi)\|^2,$$

where $\odot$ is the Hadamard product. $\mathbf{W}_\Lambda$ and $\mathbf{T}_\Lambda$ denote the weight and target matrices for the loading matrix. In contrast, $\mathbf{W}_\Psi$ and $\mathbf{T}_\Psi$ denote the weight and target matrices for the factor correlations. $\mathbf{T}_\Psi$ must be symmetric and $\mathbf{W}_\Psi$ must be off-diagonal symmetric with nonnegative elements. Lastly, the scalar w represents the relative contribution of the second term with respect to the former.

This new rotation criterion can effectively penalize the unwanted correlations specified as zeros in the target for the factor correlation matrix and the result can be very similar to standard target rotation with the orthoblique manifold. However, in our orthoblique approach, the roles of the manifold and rotation criteria are separated: the target criterion is used

only to orient the loading matrix, while the orthoblique manifold imposes the trait-method orthogonality exactly. Hence, orthoblique rotation can be used with any rotation criteria, not just with target criteria. Moreover, the correlations encouraged to be small in extended target rotation become exact structural zeros in orthoblique rotation. As such, a benefit of our approach is that the constant parameter w that weights the relative contribution of target loadings and target correlations is not required.

Empirical applications of the orthoblique rotation

We now illustrate the use of the orthoblique manifold with two empirical applications in which neither a fully orthogonal nor a fully oblique rotation would represent the intended factor-correlation structure. The examples come from psychopathology and personality research.

We used target rotation for the MTMM application and for two of the three HEXACO specifications; the third HEXACO specification used a nontarget criterion. The targets guide the orientation of the factors without fixing the loadings to their theoretical values. The recovered patterns may therefore agree with or depart from those expectations, and the nontarget specification provides a complementary examination of the HEXACO structure.

In the psychopathology example, we use a multitrait-multimethod (MTMM) design, where the same traits are measured using different informants. In these models, the associations between indicators may arise from the substantive traits but also from the method through which the measurements were obtained. Here, child, parent, and teacher reports constitute the different methods. EFA allows us to examine whether the expected trait structure emerges without fixing the loading pattern, whereas the orthoblique manifold keeps trait and method sources of common variance separated by imposing orthogonality between both blocks of factors.

In the personality example, the HEXACO hierarchy contains many items and factors. It is therefore plausible that some items have secondary loadings or that some facets are less clearly defined than expected. EFA allows these possibilities to be examined instead of assuming that every item belongs exclusively to its designated domain and facet. However, the distinction between general and specific factors still requires selected factor correlations to be zero. Thus, the exploratory model leaves all loadings free, whereas the orthoblique manifold constrains to

zero the correlations between general and specific factors, as well as all correlations between distinct specific factors.

Thus, both examples can be studied exploratorily while preserving the theoretically specified zero correlations between theoretically unrelated factors.

All analyses were conducted with the `latent` package and the `lefa()` function.

ASEBA. An exploratory Multitrait-multimethod model

The first application uses the MTMM data analyzed by Watts et al. (2022). The correlation matrix was obtained from the supplementary materials available at https://osf.io/hvskz/. We used the ABCD sample ($N = 2,119$), which includes child-, parent-, and teacher-reported psychopathology indicators measuring three traits: Attention Problems, Externalizing, and Internalizing. [6]

The original authors included the Attention Problems indicators in an Externalizing factor (Watts et al., 2022). This decision had a substantive motivation. Although the Achenbach System of Empirically Based Assessment (ASEBA) does not include Attention Problems in its Externalizing composite, Watts et al. (2022) included these indicators because attention-deficit/hyperactivity disorder is widely regarded as a marker of childhood Externalizing. They also referred to previous ASEBA analyses in which Attention Problems loaded strongly on an Externalizing factor. An exploratory analysis can examine whether the Attention Problems indicators are adequately captured by the Externalizing factor, as assumed in the merged specification.

Following the specification used by Watts et al. (2022), the attention-problems indicators were included with Externalizing. We therefore considered five exploratory factors: two trait factors corresponding to Externalizing and Internalizing, where the Externalizing label now includes the items of the Attention Problems factor, and three method factors corresponding to children, parents, and teachers.

Each indicator is expected to reflect both a substantive trait and a measurement method, but this allocation need not hold exactly in the data. Some indicators may show secondary loadings or may represent their designated trait or method less clearly than expected. An

[6] The original data included 56 indicators but two duplicated indicators (`ARGUc` and `ARGUp`) were identified after noticing the correlation matrix was not positive-definite. These indicators were removed from the supplied correlation matrix as we did not know to which method factor they would correspond, leaving 54 indicators available for the current analysis.

exploratory partially specified target rotation allows these possibilities to be examined without fixing non-target loadings to zero. At the same time, the MTMM structure requires trait factors to remain uncorrelated with method factors, while correlations may be allowed among trait factors and among method factors. Thus, the target criterion provides the substantive orientation of the factors, whereas the orthoblique manifold constrains all trait–method factor correlations to zero.

The two trait factors and the three method factors form two internally oblique blocks such that

$$\boldsymbol{\Psi} = \begin{pmatrix} \boldsymbol{\Psi}_T & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Psi}_M \end{pmatrix}.$$

Thus, the Externalizing-Internalizing correlation and the three correlations among method factors are estimated, whereas all six trait-method correlations are constrained to zero by the orthoblique manifold.

In the code below, factor positions 1 and 2 correspond to Externalizing and Internalizing, respectively, and positions 3, 4, and 5 correspond to Children, Parents, and Teachers. Supplying `oblique = c(2, 3)` automatically selects the orthoblique manifold: the two trait factors may correlate with each other, the three method factors may correlate with one another, and all six trait–method correlations are zero. The equivalent explicit-index specification is `oblique = list(1:2, 3:5)`. There is no need to specify `projection`, because its default already allows this block specification. The single criterion `rotation = "target"` is applied to the complete loading matrix, and `target` is the correspondingly ordered target matrix for the 54 retained indicators.

The model was fitted using unweighted least squares and target rotation as follows:

```
set.seed(2026)
R <- mtmm$ABCD$R # Extract the correlation matrix
efa <- lefa(sample.cov = R, sample.nobs = 2119,
            nfactors = 5, estimator = "uls",
            oblique = c(2, 3),
            rotation = "target", target = target)
```

The results showed that the estimated correlation between Externalizing and Internalizing was .251. The correlations were .298 between Children and Parents, .240 between Children and Teachers, and .302 between Parents and Teachers. In contrast, all six correlations between

trait and method factors were zero to numerical precision as expected by the orthoblique constraints.

In the same factor order, these reported estimates give

$$\widehat{\boldsymbol{\Psi}} = \left[\begin{array}{cc|ccc} 1 & .251 & 0 & 0 & 0 \\ .251 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & .298 & .240 \\ 0 & 0 & .298 & 1 & .302 \\ 0 & 0 & .240 & .302 & 1 \end{array}\right], \tag{20}$$

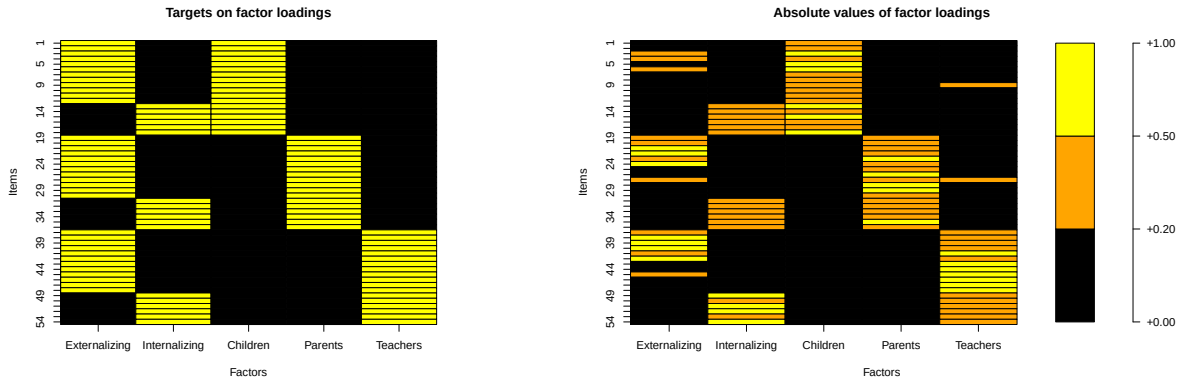
with rows and columns ordered as Externalizing, Internalizing, Children, Parents, and Teachers. The free correlations are rounded to three decimals, and the structural zeros are displayed as exact zeros.

The partially specified target rotation produced a sparse loading pattern. Among the 162 loadings targeted toward zero, 160 (98.8%) had absolute values below .20, with a median absolute value of .030. The method factors were particularly well defined: all 54 expected method loadings exceeded .20 in absolute value, with a median absolute loading of .464. In contrast, the Externalizing indicators, apart from those assessing Attention Problems, loaded poorly on the merged trait factor. Of these 18 indicators, 16 had absolute loadings below .20, with a median of .079 and a range from .028 to .259. In contrast, 15 of the 18 Attention Problems indicators exceeded .20 on the same factor, with a median absolute loading of .433. Thereby, the original Externalizing indicators contributed little to the merged trait factor, which was mainly defined by Attention Problems.

Moreover, all 18 original Externalizing indicators loaded more strongly on their corresponding method factor than on the merged trait factor, with a median absolute method loading of .530. Therefore, their weak trait loadings do not mean that they contributed little to the entire solution. Rather, their shared variation was represented more strongly by the method factors, while the merged trait factor mainly reflected Attention Problems and not Externalizing.

Overall, this exploratory solution revealed that conflating the indicators of Externalizing and Attention problems in a single factor did not recover the intended broad representation of both sets of indicators. Thus, the Externalizing label overstated the breadth of this merged factor: 15 of the 18 Attention Problems indicators, but only two of the 18 original

Externalizing indicators, had absolute trait loadings above .20.



(a) Theoretical target (b) Estimated factor loadings

Figure 6: Comparison of the theoretical and estimated factor loading patterns for the multitrait-multimethod analysis. Panel (a) shows the theoretical target structure, where yellow cells indicate items expected to load on each factor. Panel (b) displays the absolute values of the estimated factor loadings obtained through orthoblique target rotation.

From a statistical perspective, the relevant feature of this example is that rotation criteria and restrictions on the factor correlation matrix are handled separately. Target rotation determines an interpretable orientation and permits empirically relevant cross-loadings, whereas the orthoblique manifold guarantees the exact block-orthogonality structure that disentangles the variance explained by traits and methods. A conventional orthogonal rotation would have forced the correlations within the trait and method blocks to zero, while a conventional oblique rotation would have estimated the six trait-method correlations that must be excluded by the MTMM design. In both cases, the results would have been inconsistent with the theoretical model. Thus, the orthoblique manifold combined with target rotation made possible to test with EFA the adequacy of this MTMM analysis, revealing a possible modeling misspecification in Watts et al. (2022).

HEXACO. A hierarchical model with multiple oblique general factors

The second application uses the `icases` data obtained from the repository accompanying Anglim et al. (2022) at https://osf.io/72zp3/. We selected the MOOC sample ($N = 4,286$), an international sample of participants who completed the HEXACO-100. The original study reports that the sample included participants from 131 countries, with 69% being female,

and had a mean age of 34.7 years. For the present analysis, we used 96 HEXACO items corresponding to the six personality domains and their 24 facets, excluding the last four altruism items that do not belong to the HEXACO hierarchy (i.e, they are not expected to be nested in a general factor).

We fitted a 30-factor EFA model with six general factors, one for each HEXACO domain, and 24 specific factors. The first six factors were specified as one oblique block and the 24 specific factors were left orthogonal. Before any permutation used to align the factors for reporting, the specified factor correlation matrix has the form

$$\boldsymbol{\Psi} = \begin{pmatrix} \boldsymbol{\Psi}_G & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{24} \end{pmatrix},$$

where $\boldsymbol{\Psi}_G$ contains the correlations among the six domain-general factors. Therefore, the general factors can correlate, whereas every specific factor is orthogonal to the general factors and to all other specific factors. In this way, common variance would be partitioned between specific factors, representing narrow facet-level variance, and general factors, representing broader variance shared across facets.

This formulation extends the exploratory bifactor approach of Jennrich and Bentler (2011; 2012) to the case of multiple correlated general factors. In their biquartimin and bigeomin criteria, a single general-factor column is excluded from the simplicity rotation criterion, which is applied to the group-factor columns only. Here, there are six general-factor columns, and their orientation also has to be determined. We therefore applied a criterion to the general factor columns and another to the specific factor columns. Thus this example also illustrates the possibility of combining different rotation criteria within the same analysis.

To examine the extent to which the theoretical HEXACO structure is supported, we compared three rotation specifications. With 30 factors, a nontarget rotation has a large rotational search space and hence a greater risk of capitalizing on chance. Conversely, targeting a large number of loadings toward zero gives the theoretical structure a stronger role in determining factor orientation. We therefore applied geomin or oblimin to the six general factors combined with target rotation of the 24 specific factors, and compared these two solutions with oblimin applied to both blocks without a target. In the two target-based solutions, only the specific factors were targeted, providing theoretical anchoring where rotational complexity was greatest while leaving the general factors exploratory. The nontarget

oblimin specification allows both levels to be oriented by the complexity criterion and can be viewed as an extension of biquartimin rotation to multiple general factors.[7]

The three solutions recovered different aspects of the theoretical HEXACO hierarchy. In both target-based solutions, all 96 items had their largest absolute loading among the 24 specific factors on the expected facet. Without a target, this occurred for 75 of 96 items (78.1%) under oblimin. Thus, target rotation provided clearer facet separation, although the nontarget oblimin solution also recovered much of the intended facet structure without prespecifying it.

In all three solutions, the general-factor loading patterns broadly reflected the HEXACO domains. Emotionality, Extraversion, Conscientiousness, Openness, and Agreeableness were recognizable, whereas Honesty–Humility was less consistently separated in the target-based specifications. Some Honesty–Humility items showed salient cross-loadings on the Conscientiousness general factor, and additional general-factor cross-loadings remained. Because none of these specifications targeted the general-factor loadings, this correspondence was not imposed directly by the target, although targeting the specific factors could still alter the partitioning of common variance.

Figure 7 compares the theoretical loading structure (a) with the three rotated loading solutions: (b) geomin + target, (c) oblimin + target, and (d) oblimin + oblimin. The loading heatmaps display absolute values. Figure 8 presents the corresponding 6×6 general-factor correlation matrices, in panels (a), (b), and (c), respectively.

The following code illustrates the oblimin + target specification. Oblimin is applied to the six general factors and target rotation to the 24 specific factors. The argument `oblique = 6` automatically selects the orthoblique manifold and places the first six factors in one oblique block; all remaining factors are mutually orthogonal and orthogonal to the general factors. The same `oblique` specification is used in all three analyses, independently of their rotation criteria. Because the items are ordinal, the model is estimated with diagonally weighted least squares using polychoric correlations.

```
rotation_model <- list(
  oblimin = list(gamma = 0, factors = 1:6),
  target  = list(target = target[, 7:30], factors = 7:30))
```

[7] In both oblimin specifications, we set $\gamma = 0$ (quartimin). In the geomin + target specification, we set $\epsilon = .01$. For target rotation, the facet memberships determined the expected specific-factor loadings.

```
efa <- lefa(data = mooc, nfactors = 30,
            ordered = TRUE, estimator = "dwls",
            oblique = 6,
            rotation = rotation_model)
```

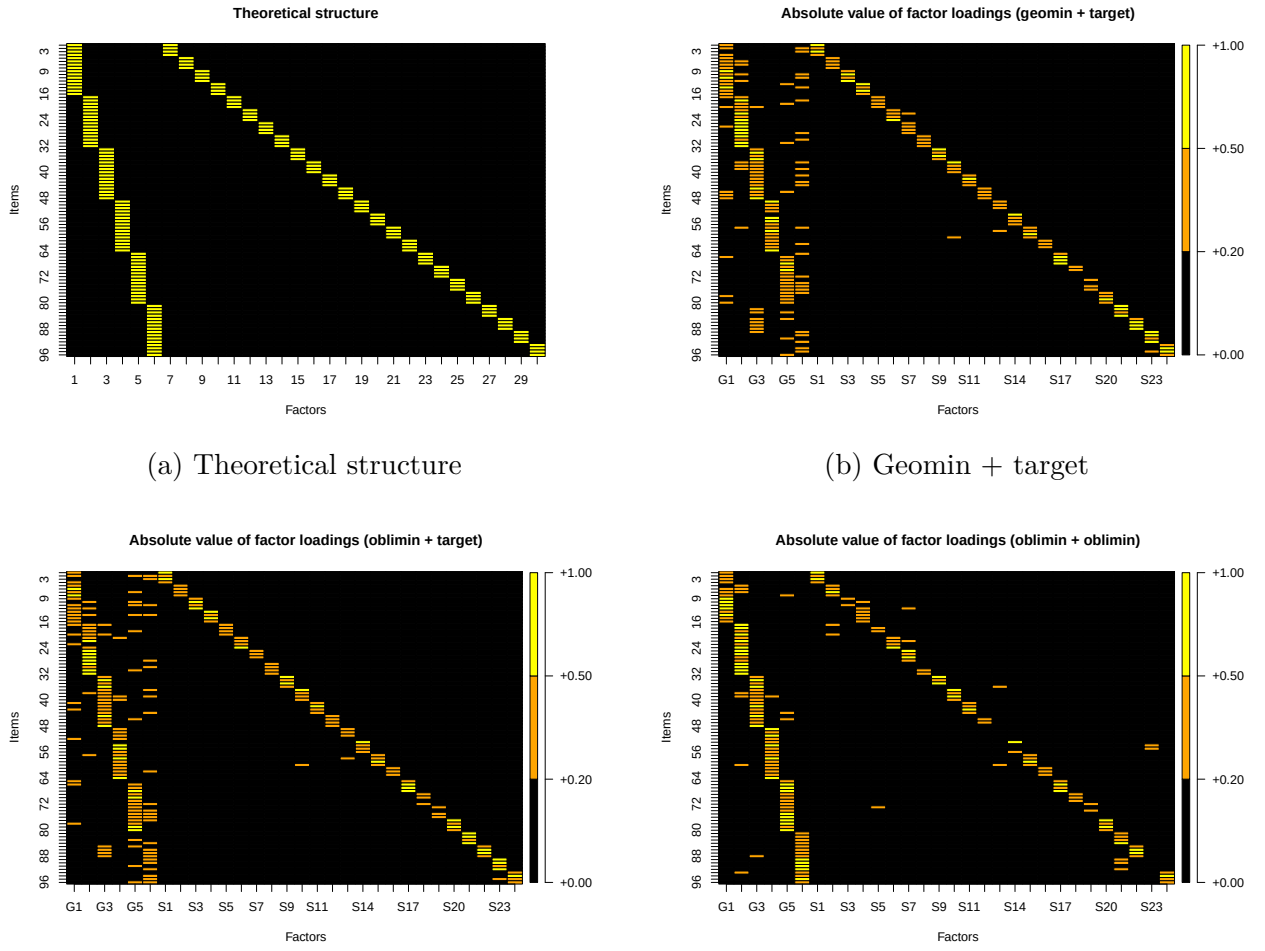
The oblimin + target solution yielded absolute correlations among the six general factors ranging from .007 to .217, with a median of .107 (Figure 8b). All general-specific correlations and all off-diagonal correlations among the 24 specific factors were zero to numerical precision, as expected by the orthoblique constraints. Therefore, of the $30(29)/2 = 435$ possible factor correlations, only the $6(5)/2 = 15$ correlations among the general factors were estimated, while the remaining 420 were fixed to zero and recovered to numerical precision.

The six general factors remained substantially exploratory in this analysis. The median absolute loading on the theoretically-expected general factor was .384; 85 of the 96 designated general-factor loadings exceeded .20 in absolute value, and 430 of the 480 general-factor cross-loadings (89.6%) were below .20. For the 24 specific factors, the median absolute theoretically expected loading was .431, and 91 of these 96 loadings exceeded .20. In contrast, 2,205 of the 2,208 specific-factor cross-loadings (99.86%) were below .20, and no specific-factor cross-loading exceeded .30.

With geomin + target (Figures 7b and 8a), the expected specific-factor loadings had a median absolute value of .427, with 91 of 96 exceeding .20; 2,204 of the 2,208 cross-loadings on other specific factors were below .20. Thus, geomin and oblimin produced comparably sparse specific-factor patterns when paired with target rotation. For the general factors, the median absolute designated loading was .414, 82 of the 96 designated loadings exceeded .20, and 426 of the 480 general-factor cross-loadings were below .20. The absolute general-factor correlations ranged from .007 to .214 (median .116). Compared with oblimin + target, the differences were modest, especially for the specific-factor block.

The oblimin + oblimin solution (Figures 7d and 8c) recovered the expected specific factor for 75 items. Its median absolute expected specific-factor loading was .348, with 72 of 96 exceeding .20; 2,193 of the 2,208 specific-factor cross-loadings were below .20, and five exceeded .30. In contrast, its general-factor block showed a clearer simple structure than in the two target-based solutions: the median absolute expected general-factor loading was .459, 95 of 96 exceeded .20, and 468 of the 480 general-factor cross-loadings were below .20. The absolute correlations among the general factors ranged from .015 to .276 (median .138), larger at the

upper end than under the two target-based solutions. This configuration therefore favored a clearer general-factor structure at the expense of less distinct specific factors.



(a) Theoretical structure

(b) Geomin + target

(c) Oblimin + target

(d) Oblimin + oblimin

Figure 7: Theoretical loading structure and absolute values of estimated factor loadings for the HEXACO analyses: (a) theoretical structure, (b) geomin + target, (c) oblimin + target, and (d) oblimin + oblimin. The two target-based specifications use target rotation for the specific factors, whereas the nontarget specification uses oblimin for both general and specific factors.

The conceptual advantage of the orthoblique manifold was particularly clear in this example. A fully oblique 30-factor rotation would allow all 435 factor correlations, although only the 15 correlations among the six general factors are needed for this representation. Conversely, a fully orthogonal rotation would force these 15 correlations to zero. In the two target-based specifications, the general-factor orientations remain determined by a simplicity criterion, while target rotation anchors the 24 specific factors where the risk of capitalizing on chance is especially high. The nontarget oblimin specification provides a comparison without this

	G_1	G_2	G_3	G_4	G_5	G_6
G_1	1.00	0.15	0.06	−0.09	0.13	−0.21
G_2	0.15	1.00	0.03	−0.12	0.13	−0.01
G_3	0.06	0.03	1.00	−0.19	0.07	−0.14
G_4	−0.09	−0.12	−0.19	1.00	−0.12	0.01
G_5	0.13	0.13	0.07	−0.12	1.00	−0.02
G_6	−0.21	−0.01	−0.14	0.01	−0.02	1.00

(a) Geomin + target

	G_1	G_2	G_3	G_4	G_5	G_6
G_1	1.00	−0.11	−0.12	0.08	−0.22	0.05
G_2	−0.11	1.00	0.08	−0.12	0.11	−0.01
G_3	−0.12	0.08	1.00	−0.20	0.10	−0.11
G_4	0.08	−0.12	−0.20	1.00	−0.13	0.04
G_5	−0.22	0.11	0.10	−0.13	1.00	−0.11
G_6	0.05	−0.01	−0.11	0.04	−0.11	1.00

(b) Oblimin + target

	G_1	G_2	G_3	G_4	G_5	G_6
G_1	1.00	−0.02	−0.10	0.06	0.25	0.04
G_2	−0.02	1.00	0.19	−0.18	−0.28	−0.01
G_3	−0.10	0.19	1.00	−0.14	−0.07	−0.18
G_4	0.06	−0.18	−0.14	1.00	0.08	0.15
G_5	0.25	−0.28	−0.07	0.08	1.00	0.24
G_6	0.04	−0.01	−0.18	0.15	0.24	1.00

(c) Oblimin + oblimin

Figure 8: Factor correlation matrices among the six general HEXACO factors for (a) geomin + target, (b) oblimin + target, and (c) oblimin + oblimin. All entries are rounded to two decimal places. Correlations between general and group factors and between distinct group factors are constrained to zero and thus are not displayed.

theoretical anchoring.

The three solutions illustrate a trade-off between the general and specific levels of the HEXACO hierarchy. Both target-based solutions yielded highly interpretable facet structures while leaving general-factor loadings exploratory. The nontarget oblimin solution provided a particularly simple general-factor pattern but weaker facet recovery. As these results come from a single sample, they should not be interpreted as a general ranking of the rotation criteria.

Taken together, these applications illustrate the distinction between rotation criteria and manifold restrictions. Complexity criteria can yield different sparsity and parsimony patterns for each block of factors whereas the orthoblique manifold defines the admissible factor-correlation geometry exactly. The two mechanisms can therefore be combined without turning the factor model into a confirmatory simple-structure model.

Discussion

Conventional factor rotation has treated factors as either completely orthogonal or completely oblique. We generalized this framework by introducing the orthoblique rotation, which allows factors to correlate within specified blocks while remaining orthogonal between blocks. This new rotation approach accommodates theoretically motivated or design-imposed constraints on the factor correlation matrix and hence expands the flexibility and modeling capabilities of EFA. In this sense, orthoblique rotation helps bridge the gap between exploratory and confirmatory factor analysis by allowing researchers to impose selected factor-correlation constraints while retaining the exploratory estimation of all factor loadings. Thus, prior theoretical knowledge can be incorporated into the factor model without imposing a confirmatory loading structure, preserving the ability of EFA to uncover unexpected cross-loadings and departures from theoretical expectations.

To develop the orthoblique geometry, we provided a detailed and self-contained review of the GPA for orthogonal and oblique rotations, describing and connecting many mathematical concepts and visual representations that are often overlooked in discussions concerning factor rotation. In particular, starting from the defining constraints of the orthoblique manifold, we derived its tangent and normal spaces to find an expression for the Riemannian gradient and also designed a suitable retraction algorithm to calculate appropriate updates for the rotation matrix in the GPA. We hope that these explicit connections make factor rotation a less obscure matter in EFA and help researchers understand how further extensions can be constructed. We also showed that the conventional orthogonal and oblique rotation geometries emerge as special cases of the orthoblique framework, establishing orthoblique rotation as a natural generalization of both approaches.

The empirical applications illustrated two complementary uses of this new orthoblique geometry: an MTMM representation in which trait and method factors can correlate within their respective blocks while remaining orthogonal between blocks, and a bifactor model in which several general factors may correlate among them while remaining orthogonal to specific factors that are also mutually orthogonal. As such, the present work not only builds on the general-purpose rotation framework developed by Jennrich (2001), Jennrich (2002), and Bernaards and Jennrich (2005). It also extends the exploratory bifactor rotation methods based on biquartimin and bigeomin (Jennrich & Bentler, 2011, 2012) to account for scenarios involving multiple general factors.

Additionally, we provided a new R package to exploit these new rotation capabilities. The function `lrotate()` implements orthogonal, oblique, and orthoblique rotations and also allows combinations of different complexity criteria. Through the `projection`, `oblique`, and `rotation` arguments, researchers can specify a desired factor-correlation pattern with structural zeros and different complexity criteria for specific columns of the loading matrix. As a result, combining orthoblique rotation with multiple rotation criteria substantially expands the range of possible EFA specifications and opens up new opportunities for developing innovative approaches to exploring factor structures. The bifactor application with multiple general factors illustrated only a few of these combinations; we have not examined the many other possibilities or established which will be most useful in different substantive settings. The `latent` package makes these unexplored combinations already available to researchers, who can investigate new rotation configurations based on their parameter recovery, stability, and interpretability.

In this sense, the current work provides both a geometric background and a software basis for further research on factor rotation and, more broadly, on EFA. However, an important limitation concerns the identification and ordering of factors in EFA. Factor solutions are identified only up to sign changes and permutations of the factors. Consequently, the factor positions specified as oblique blocks may not correspond to the substantive factors that researchers intended to allow to correlate. It is therefore important to inspect the rotated loading matrix to verify that the factors assigned to each oblique block correspond to the intended factor structure. Another limitation is that we did not assess the proclivity of orthoblique rotation to local minima solutions. In factor rotation, different starting values for the rotation matrix may lead to different stationary points, potentially yielding different loading and factor correlation patterns. Although the `latent` package supports conducting rotation from multiple random starting values, these do not guarantee convergence to the global minimum. Furthermore, the accuracy and stability of parameter recovery have not yet been systematically evaluated under different factor structures. Future simulation studies should therefore examine the robustness of orthoblique rotation to different starting values, rotation criteria, and population factor structures.

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